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PURE MATHEMATICS WITH STATISTICS • UPPER SIXTH

UPPER SIXTH — PURE MATHEMATICS WITH STATISTICS

CHAPTER 16

HYPOTHESIS TESTING

SUGGESTED TIME: 10 PERIODS

IN THIS CHAPTER

This chapter introduces the logic of statistical hypothesis testing: formulating null and alternative hypotheses, choosing a significance level, and using critical regions to make decisions about population parameters. Students learn to carry out tests for a population mean and for the difference between two means, and to interpret Type I and Type II errors in real Cameroonian contexts.

- By the end of this chapter I can state a null hypothesis H0 and an alternative hypothesis H1 appropriate to a given problem.
- By the end of this chapter I can distinguish between one-tailed and two-tailed tests and justify the choice.
- By the end of this chapter I can explain the level of significance and compute the appropriate test statistic.
- By the end of this chapter I can determine critical values and the critical region for a given test.
- By the end of this chapter I can describe Type I and Type II errors and their probabilities.
- By the end of this chapter I can apply the full five-step hypothesis testing procedure.

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THIS SIGNED COURSE CONTAINS

COURSE — complete, illustrated, step by step

METHODS — and worked examples

EXERCISES — exam style + solutions

SUMMARY SHEET — the essentials to remember

CONTENTS

1. Statistical Hypotheses and Types of Tests
2. Significance Level, Test Statistics and Critical Regions
3. Errors in Hypothesis Testing and the Testing Procedure
4. Hypothesis Test for a Population Mean
5. Test for the Difference Between Two Means (Large Samples)
6. Applications, Consolidation and Exam Practice
7. Integration activity
8. Methods and worked examples
9. Exercises
10. Solutions to the exercises
11. Summary sheet

CHAPTER OBJECTIVES

- By the end of this chapter I can state a null hypothesis H0 and an alternative hypothesis H1 appropriate to a given problem.
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- By the end of this chapter I can explain the level of significance and compute the appropriate test statistic.
- By the end of this chapter I can determine critical values and the critical region for a given test.
- By the end of this chapter I can describe Type I and Type II errors and their probabilities.
- By the end of this chapter I can apply the full five-step hypothesis testing procedure.
- By the end of this chapter I can test a hypothesis about a population mean using a sample (normal distribution, known or estimated variance).
- By the end of this chapter I can test for a difference between two population means using large samples.
- By the end of this chapter I can interpret the conclusion of a test in everyday language and apply testing to real-life situations.

PREREQUISITES

- The normal distribution and use of standard normal tables (Lower Sixth).
- Sampling distributions: the sample mean $\bar{X} \sim N(\mu, \sigma^2/n)$ and the Central Limit Theorem.
- Confidence intervals for a population mean.
- Basic probability: complementary events and conditional reasoning.
- Estimators: unbiased estimates of μ and σ^2 from a sample.

1 STATISTICAL HYPOTHESES AND TYPES OF TESTS

A bottling company in Douala claims its Top Ananas juice bottles contain exactly 50 cl on average, but a consumer association suspects the machines are under-filling them. Similarly, a Yaoundé hospital wants to know whether a new malaria treatment really lowers fever faster than the standard one, and the GCE Board wonders whether this year's candidates performed better than last year's. In each case, we cannot check every bottle, every patient or every script — we can only take a sample. How do we then decide, with a controlled risk of being wrong, whether the claim should be rejected? This is exactly the science of **hypothesis testing**.

1.1 What is a Statistical Hypothesis?

DEFINITION **Statistical Hypothesis**

A statistical hypothesis is a claim or statement about one or more parameters of a population distribution (e.g. the mean μ, the proportion p, the variance σ^2). It is an assumption we make about the population that we wish to test using sample data.

In the Top Ananas example, the parameter of interest is the population mean volume μ (in cl). The company's claim is the hypothesis $\mu = 50$. The consumer association's suspicion is $\mu < 50$. Both are statistical hypotheses because they make a statement about the population mean.

PROPERTY **Simple vs. Composite Hypotheses**

- A simple hypothesis completely specifies the population distribution (e.g. $H_0 : \mu = 50$ when σ^2 is known).
- A composite hypothesis does not completely specify the distribution; it contains a range of values (e.g. $H_1 : \mu < 50$ or $H_1 : \mu \neq 50$).

In practice, the null hypothesis is often simple (a single value) while the alternative is composite.

1.2 Null and Alternative Hypotheses

Every hypothesis test pits two competing hypotheses against each other.

DEFINITION **Null Hypothesis H_0**

The null hypothesis, denoted H_0, is the hypothesis that is initially assumed to be true. It represents the status quo, the established fact, or the claim to be challenged. It always contains an equality sign ($=$, $\leq$, or $\geq$).

DEFINITION **Alternative Hypothesis H_1**

The alternative hypothesis, denoted H_1 (or H_a), is the hypothesis that we suspect might be true instead of H_0. It represents the research claim, the change, or the effect we are looking for. It never contains an equality sign; it uses $\neq$, $<$, or $>$.

WARNING **The Equality Rule — Examinable**

H_0 must always contain the equality sign. Write $H_0 : \mu = \mu_0$ (or $\mu \leq \mu_0$, $\mu \geq \mu_0$). Never write $H_0 : \mu < \mu_0$ or $H_0 : \mu > \mu_0$. The test statistic's distribution is derived assuming the boundary value (usually equality) is true.

WARNING **We Never "Prove" H_0**

A hypothesis test can only provide evidence against H_0. If the sample data are inconsistent with H_0, we reject H_0 in favour of H_1. If the data are not sufficiently inconsistent, we fail to reject H_0. We never say "accept H_0" or "H_0 is true". Absence of evidence against H_0 is not evidence for H_0.

1.3 One-Tailed and Two-Tailed Tests

The form of H_1 determines the **type of test** and the location of the **critical region** (rejection region) under the sampling distribution of the test statistic.

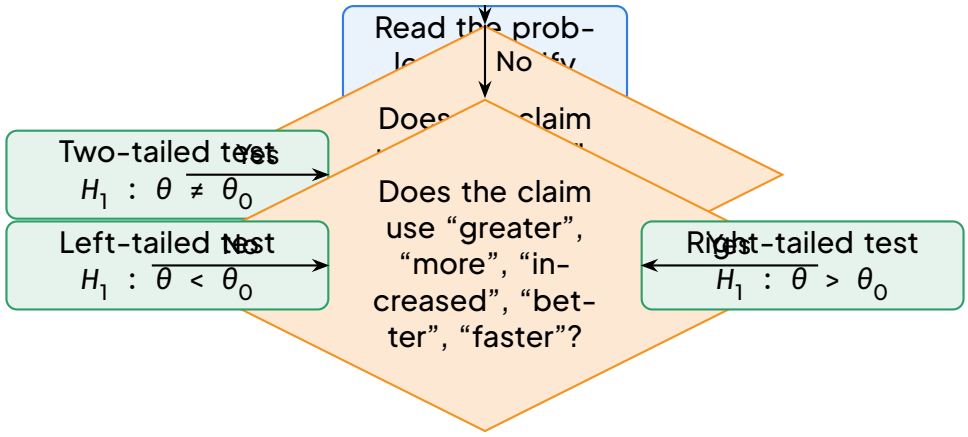
PROPERTY **Forms of H_1 and Corresponding Tests**

Let the parameter be θ (e.g. μ, p) and the hypothesised value be θ_0.

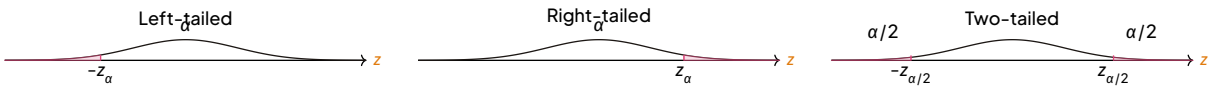
Alternative H_1	Name of Test	Critical Region Location	Typical Wording
$H_1 : \theta < \theta_0$	Left-tailed (lower-tailed)	Left tail only	"less than", "decreased", "lower than", "under-filling"
$H_1 : \theta > \theta_0$	Right-tailed (upper-tailed)	Right tail only	"greater than", "increased", "higher than", "better than", "faster than"
$H_1 : \theta \neq \theta_0$	Two-tailed	Both tails (split equally)	"different from", "changed", "not equal to", "whether ... differs"

METHOD **Choosing the Test from the Problem Statement**

1. Identify the parameter (μ, p, $\mu_1 - \mu_2$, etc.) and define it clearly in a sentence.
2. Locate the claim or research question. That claim usually becomes H_1.
3. Look for keywords:
   - "different from", "changed", "not equal" → Two-tailed ($H_1 : \theta \neq \theta_0$).
   - "greater than", "more than", "increased", "better", "faster" → Right-tailed ($H_1 : \theta > \theta_0$).
   - "less than", "fewer than", "decreased", "lower", "under" → Left-tailed ($H_1 : \theta < \theta_0$).
4. Write H_0 as the complementary statement including the equality at the boundary.



Flowchart: from the wording of the problem to the form of H_1.



Rejection regions (shaded) for the three types of tests at significance level α (standard normal distribution shown).

1.4 Significance Level, Test Statistic and Critical Region

DEFINITION **Level of Significance α**

The level of significance, denoted α, is the maximum probability of rejecting the null hypothesis H_0 when it is actually true. It is the size of the critical region. Common values are 0.05 (5%), 0.01 (1%) and 0.10 (10%).

DEFINITION **Test Statistic**

A test statistic is a quantity calculated from the sample data (e.g. the sample mean $\bar{x}$, sample proportion $\hat{p}$) whose sampling distribution is completely known when H_0 is true. It is used to decide whether to reject H_0. For large samples testing a mean with known variance, the test statistic is $Z = \dfrac{\bar{X} - \mu_0}{\sigma/\sqrt{n}} \sim N(0,1)$ under H_0.

DEFINITION **Critical Region and Critical Values**

The critical region (or rejection region) is the set of values of the test statistic for which we reject H_0. The critical value(s) are the boundary value(s) that separate the critical region from the acceptance region. They are denoted z_α, $z_{\alpha/2}$, t_α, etc., depending on the distribution and α.

KEY POINTS **Summary of Critical Values for Standard Normal Z**

Test Type	Critical Region	Critical Value(s)	Common α Values
Left-tailed ($H_1 : \theta < \theta_0$)	$Z < -z_\alpha$	$-z_\alpha$	$\alpha = 0.05 \Rightarrow -1.645$
Right-tailed ($H_1 : \theta > \theta_0$)	$Z > z_\alpha$	z_α	$\alpha = 0.05 \Rightarrow 1.645$
Two-tailed ($H_1 : \theta \neq \theta_0$)	$\lvert Z \rvert > z_{\alpha/2}$	$\pm z_{\alpha/2}$	$\alpha = 0.05 \Rightarrow \pm 1.96$

1.5 Worked Examples

WORKED EXAMPLE 1 **Formulating Hypotheses**

For each situation, define the parameter, state H_0 and H_1, and identify the type of test.

1. A manufacturer claims the mean lifetime of his light bulbs is 1000 hours. A consumer group believes the mean lifetime is actually less.
2. The pass rate in Mathematics at GCE A-Level has been 65% for years. The Ministry of Education introduces a new syllabus and wants to know if the pass rate has changed.
3. A new fertilizer is claimed to increase the mean yield of maize above the current mean of 2.5 tonnes/hectare.

SOLUTION

Solution.

1. Parameter: μ = population mean lifetime (hours).
   Claim: "less than 1000" $\rightarrow H_1 : \mu < 1000$.
   $H_0 : \mu = 1000$ (equality at boundary).
   Left-tailed test.
2. Parameter: p = population proportion of candidates who pass.
   Claim: "has changed" $\rightarrow H_1 : p \neq 0.65$.
   $H_0 : p = 0.65$.
   Two-tailed test.

3. Parameter: μ = population mean yield (tonnes/hectare).
   Claim: "increase ... above 2.5" → $H_1 : \mu > 2.5$.
   $H_0 : \mu = 2.5$.
   Right-tailed test.

WORKED EXAMPLE 2 **Top Ananas Bottling**

The Top Ananas company states that the mean volume of juice in its 50 cl bottles is 50 cl. A consumer association takes a random sample of 36 bottles and finds a sample mean of 49.7 cl with a known population standard deviation of 0.5 cl. They want to test, at the 5% significance level, whether the bottles are under-filled.

1. State the hypotheses and the type of test.
2. Explain why H_0 is $\mu = 50$ and not $\mu \geq 50$.

SOLUTION

Solution.

1. Parameter: μ = true mean volume of all Top Ananas 50 cl bottles (in cl).
   The association suspects under-filling → claim is $\mu < 50$.
   $H_1 : \mu < 50$ (Left-tailed test).
   $H_0 : \mu = 50$ (boundary value).
2. Both $H_0 : \mu = 50$ and $H_0 : \mu \geq 50$ are technically correct null hypotheses for a left-tailed test because the test statistic's distribution is derived using the boundary value $\mu = 50$. However, the convention (and GCE requirement) is to write the simple form $H_0 : \mu = 50$. Writing $H_0 : \mu \geq 50$ is a composite null and is not standard practice at this level; it can lead to confusion about which value to use for the test statistic calculation. Always write H_0 with $=$.

1.6 GCE Examination Tips

KEY POINTS **Key Points for Section 1**

- Define the parameter in words before writing symbols (e.g. "Let μ be the mean volume in cl of all bottles produced by the machine").
- H_0 always contains $=$ (or $\leq$, $\geq$). H_1 never contains $=$.
- The claim you are trying to prove goes into H_1.
- Keywords → Test type:

- "different / changed / not equal" → Two-tailed.
- "greater / more / increased / better / faster" → Right-tailed.
- "less / fewer / decreased / lower / under" → Left-tailed.

- Conclusion language: "Reject H_0" or "Fail to reject H_0". Never "Accept H_0".

WARNING **Common Mistakes in GCE Papers**

- Writing $H_0 : \mu < 50$ and $H_1 : \mu = 50$ (swapped).
- Writing $H_0 : \mu \neq 50$ (equality missing).
- Not defining the parameter μ or p clearly.
- Saying "Accept H_0" or "H_0 is true".
- Choosing a two-tailed test when the problem says "greater than" (or vice versa).
- Confusing the critical value (z_α) with the calculated test statistic (z_{calc}).

DID YOU KNOW? **Historical Note**

The formal framework of hypothesis testing was developed in the 1920s–1930s by Jerzy Neyman and Egon Pearson (building on earlier work by R.A. Fisher). They introduced the concepts of the null hypothesis, alternative hypothesis, Type I/II errors, and the critical region. Their work was motivated by industrial quality control — exactly the kind of problem faced by the Top Ananas bottling plant!

EXERCISE 1 **Practice: Formulating Hypotheses**

For each scenario, define the parameter, write H_0 and H_1, and state the type of test.

1. A pharmaceutical company claims its new painkiller relieves pain in less than 20 minutes on average. The current drug takes 20 minutes.
2. The mean weight of 1 kg bags of rice from a factory is supposed to be 1 kg. Quality control wants to check if the packing machine needs recalibration (i.e. if the mean differs from 1 kg).
3. A politician claims that more than 60% of voters in his region support him. An opponent suspects the true support is lower.

4. A school principal believes the mean score of her students in the GCE Mathematics paper this year is higher than last year's mean of 11.5/20.

SOLUTION **Exercise 1**

1. μ = mean relief time (min). $H_0 : \mu = 20$, $H_1 : \mu < 20$. Left-tailed.
2. μ = mean weight (kg). $H_0 : \mu = 1$, $H_1 : \mu \neq 1$. Two-tailed.
3. p = proportion of voters supporting the politician. The claim to be tested (the challenger's suspicion) is that support is lower than claimed. $H_0 : p = 0.60$, $H_1 : p < 0.60$. Left-tailed. (Note: If we were testing the politician's claim directly, H_1 would be $p > 0.60$; context determines which claim is the "research hypothesis" H_1.)
4. μ = mean score this year. $H_0 : \mu = 11.5$, $H_1 : \mu > 11.5$. Right-tailed.

2 SIGNIFICANCE LEVEL, TEST STATISTICS AND CRITICAL REGIONS

In the previous section we learnt how to state the **null hypothesis** H_0 and the **alternative hypothesis** H_1, and how to classify a test as **one-tailed** or **two-tailed**. But stating hypotheses is not enough: we must now answer the practical question "how do we actually decide?". A sample mean $\bar{x}$ will almost never equal μ_0 exactly, even when H_0 is true, because of sampling variability. So we need a precise rule that tells us when the observed sample is too far from what H_0 predicts to be explained by chance alone. That rule is built on three pillars: the **level of significance** α, the **test statistic**, and the **critical region**.

2.1 The Level of Significance α

Intuition.

Suppose a miller in Bamenda claims that his bags of flour have mean mass $\mu_0 = 50$ kg. You weigh 36 bags and find $\bar{x} = 49.4$ kg. Is the difference of 0.6 kg just bad luck, or evidence that the bags are underfilled? There is always some chance that a genuine 50 kg process produces $\bar{x} = 49.4$ kg. Before collecting data, we must decide how small that chance must be before we stop believing H_0. That threshold probability is the level of significance.

DEFINITION **Level of significance**

The **level of significance** of a test, denoted α, is the probability, fixed in advance, of re-

jecting H_0 when H_0 is actually true:

$$\alpha = P(\text{reject } H_0 \mid H_0 \text{ true}).$$

The most commonly used levels are $\alpha = 0.10$ (10%), $\alpha = 0.05$ (5%) and $\alpha = 0.01$ (1%). The test is then called a "test at the 5% level", etc.

Choosing $\alpha = 0.05$ means: "if H_0 is true, I accept a 5% risk of wrongly condemning it." A smaller α (say 1%) makes the test more cautious — we demand stronger evidence before rejecting H_0 — but makes it harder to detect a real difference. In the GCE examination the level is always given in the question; you never have to choose it yourself.

KEY POINTS **Key point**

α is a probability fixed before looking at the data. It measures the risk of a wrong rejection of H_0, never the probability that H_0 is true.

2.2 The Test Statistic and its Distribution under H_0

Intuition.

To compare the sample with H_0 we cannot use $\bar{x}$ directly, because "far from μ_0" depends on the units and on the spread of the data. A difference of 0.6 kg is enormous if bags vary by grams, negligible if they vary by kilograms. The remedy is to standardise: measure the gap $\bar{x} - \mu_0$ in units of the standard error $\sigma/\sqrt{n}$.

DEFINITION **Test statistic**

A **test statistic** is a quantity computed from the sample whose probability distribution is completely known when H_0 is true. For tests on a population mean with known variance σ^2 (or with large sample size $n \geq 30$ where the sample standard deviation s is used as an estimate of σ), the test statistic is the standardised sample mean:

$$Z = \frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}} \quad \text{or} \quad Z = \frac{\bar{X} - \mu_0}{s/\sqrt{n}} \ (n \text{ large}).$$

PROPERTY **Distribution of the test statistic under H_0**

If $H_0 : \mu = \mu_0$ is true, and either the population is $N(\mu_0, \sigma^2)$ or the sample size n is large

(Central Limit Theorem), then

$$Z = \frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}} \sim N(0,1).$$

This result is examinable: it follows from $\bar{X} \sim N\left(\mu_0, \frac{\sigma^2}{n}\right)$ and the standardisation formula $Z = \frac{\bar{X} - E(\bar{X})}{\sqrt{\mathrm{Var}(\bar{X})}}$.

Because the distribution of Z under H_0 is the standard normal, we can compute exactly how surprising any observed value z is, using the normal tables provided in the examination.

EXEMPLE **Computing a test statistic**

A machine fills packets of sugar with $\sigma = 8$ g known from long experience. We test $H_0 : \mu = 500$ g. A sample of $n = 64$ packets gives $\bar{x} = 497$ g. Then

$$z = \frac{497 - 500}{8/\sqrt{64}} = \frac{-3}{1} = -3.$$

The observed sample mean lies 3 standard errors below μ_0 — a rare event under H_0.

2.3 Critical Values from the Standard Normal Table

The **critical value** is the boundary on the Z-scale beyond which we decide the result is too extreme for H_0. It is read from the standard normal tables so that the tail area equals α (one-tailed) or $\alpha/2$ in each tail (two-tailed).

PROPERTY **Critical values of $N(0,1)$ at the usual levels**

Level α	10%	5%	1%
One-tailed test	1.282	1.645	2.326
Two-tailed test	1.645	1.960	2.576

More precisely: one-tailed critical values are $z_{0.10} = 1.282$, $z_{0.05} = 1.645$, $z_{0.01} = 2.326$; two-tailed critical values are $z_{0.05} = 1.645$, $z_{0.025} = 1.960$, $z_{0.005} = 2.576$, where z_p denotes the value with $P(Z > z_p) = p$.

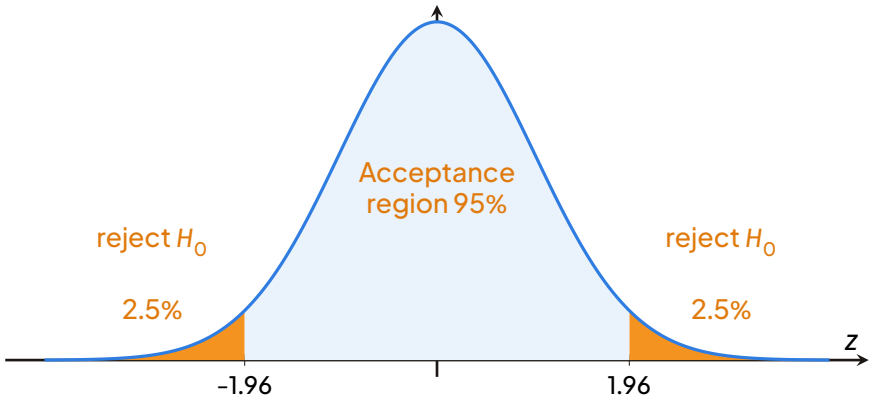
Where do these numbers come from?

For a two-tailed test at 5%, we need z with $P(Z > z) = 0.025$, i.e. $\Phi(z) = 0.975$. Reading the normal table in reverse gives $z = 1.96$. For a one-tailed test at 5% we need $\Phi(z) = 0.95$, giving

$z = 1.645$. The value 2.576 comes from $\Phi(z) = 0.995$, and 2.326 from $\Phi(z) = 0.99$.

WARNING **The most frequent GCE mistake**

For a two-tailed test at the 5% level, the total tail probability 5% is shared equally between the two tails: each tail carries 2.5%. The critical values are therefore ±1.96, not ±1.645. The value 1.645 belongs to the one-tailed 5% test. Always ask yourself first: "one tail or two tails?"



Two-tailed test at the 5% level: critical region $|z| > 1.96$ (shaded tails, 2.5% each) and acceptance region $-1.96 \le z \le 1.96$.

2.4 Critical Region, Acceptance Region and Boundary Values of $\bar{X}$

DEFINITION **Critical and acceptance regions**

The **critical region** (or rejection region) C is the set of values of the test statistic for which H_0 is rejected; it satisfies $P(Z \in C \mid H_0 \text{ true}) = \alpha$. Its complement is the **acceptance region**. The boundary points of C are the **critical values**.

The shape of C depends on H_1:

- $H_1 : \mu > \mu_0$ (right tail): $C = \{z > z_\alpha\}$;
- $H_1 : \mu < \mu_0$ (left tail): $C = \{z < -z_\alpha\}$;
- $H_1 : \mu \neq \mu_0$ (two tails): $C = \{|z| > z_{\alpha/2}\}$.

Sometimes the examiner asks for the critical region in terms of $\bar{X}$ itself. Since $Z = \dfrac{\bar{X} - \mu_0}{\sigma/\sqrt{n}}$, the

inequality $Z > z_\alpha$ is equivalent to

$$\bar{X} > \mu_0 + z_\alpha \frac{\sigma}{\sqrt{n}}.$$

The number $\mu_0 + z_\alpha \frac{\sigma}{\sqrt{n}}$ is the **boundary value of** $\bar{X}$.

WARNING **Critical value versus boundary value**

Do not confuse the critical value (a number on the standardised Z-scale, e.g. 1.645) with the boundary value of $\bar{X}$ (a number in the original units, e.g. 50.82 kg). Both describe the same borderline, but on different scales. Read the question carefully: "find the critical region for the sample mean" requires the $\bar{X}$ scale.

METHOD **Finding the critical region for a given α and type of test**

1. Identify the type of test from H_1: right-tailed ($>$), left-tailed ($<$) or two-tailed ($\neq$).
2. Choose the tail probability: α in one tail, or $\alpha/2$ in each tail for a two-tailed test.
3. Read the critical value(s) from the normal table: $\Phi(z) = 1 - \alpha$ (one tail) or $\Phi(z) = 1 - \alpha/2$ (two tails).
4. Write the critical region on the Z-scale, with the correct sign(s).
5. If required, convert to the $\bar{X}$-scale using $\bar{X} = \mu_0 + z\frac{\sigma}{\sqrt{n}}$.

WORKED EXAMPLE 3 **Critical region on both scales**

The lifetime of bulbs produced at a factory in Douala is known to have standard deviation $\sigma = 120$ hours. We test $H_0 : \mu = 1500$ hours against $H_1 : \mu < 1500$ hours at the 5% level, using a sample of $n = 36$ bulbs. Find the critical region (a) for Z; (b) for $\bar{X}$.

SOLUTION

Solution. The test is left-tailed at $\alpha = 0.05$.
(a) We need z with $\Phi(z) = 0.95$, i.e. $z = 1.645$. The critical region is

$$C = \{z < -1.645\}.$$

(b) The standard error is $\frac{\sigma}{\sqrt{n}} = \frac{120}{\sqrt{36}} = \frac{120}{6} = 20$ hours. The boundary value of $\bar{X}$ is

$$\mu_0 - z_\alpha \frac{\sigma}{\sqrt{n}} = 1500 - 1.645 \times 20 = 1500 - 32.9 = 1467.1 \text{ hours}.$$

Hence the critical region is $\bar{x} < 1467.1$ hours: if the sample mean lifetime falls below 1467.1 hours, we reject H_0 at the 5% level.

2.5 The Decision Rule and the p-value

DEFINITION **Decision rule**

Compute the observed value z of the test statistic from the sample. Reject H_0 at the level α if z falls in the critical region; otherwise do not reject H_0.

A useful extra, increasingly appreciated by GCE examiners, is the **p-value**.

DEFINITION **p-value**

The **p-value** is the probability, computed assuming H_0 true, of obtaining a result at least as extreme as the one observed. If the p-value is less than α, we reject H_0.

For a two-tailed test, "at least as extreme" means in either tail, so the p-value is twice the one-tail probability. For a one-tailed test, it is the probability in the direction specified by H_1.

EXEMPLE **Calculating p-values**

Suppose the observed test statistic is $z = -2.1$.

- For a two-tailed test: p-value $= 2 \times P(Z < -2.1) = 2 \times 0.0179 = 0.0358$.
- For a left-tailed test ($H_1 : \mu < \mu_0$): p-value $= P(Z < -2.1) = 0.0179$.
- For a right-tailed test ($H_1 : \mu > \mu_0$): p-value $= P(Z > -2.1) = 1 - 0.0179 = 0.9821$.

WORKED EXAMPLE 4 **A complete decision using critical region and p-value**

Using the sugar-packet data above: $H_0 : \mu = 500$ g, $H_1 : \mu \neq 500$ g, $\sigma = 8$ g, $n = 64$, $\bar{x} = 497$ g. Test at the 5% level.

SOLUTION

Solution. Two-tailed test at 5%: critical values ± 1.96, so $C = \{|z| > 1.96\}$.

Test statistic: $z = \dfrac{497 - 500}{8/8} = -3$.

Since $-3 < -1.96$, z lies in the critical region: we reject H_0 at the 5% level. There is significant evidence that the mean mass differs from 500 g.
p-value approach: p-value $= 2 \times P(Z < -3) = 2 \times 0.00135 = 0.0027$. Since $0.0027 < 0.05$, we reject H_0. The small p-value confirms strong evidence against H_0.

EXERCISE 2 **Critical regions and decisions**

A machine produces rods with $\sigma = 0.9$ cm. Test $H_0 : \mu = 25$ cm using $n = 81$ rods.

1. Find the critical region for $\bar{X}$ for a two-tailed test at the 1% level.
2. Find the critical region for $\bar{X}$ for a right-tailed test ($H_1 : \mu > 25$) at the 5% level.
3. The sample gives $\bar{x} = 25.21$ cm. State your conclusion in each case.

SOLUTION **Exercise 1**

The standard error is $\dfrac{0.9}{\sqrt{81}} = 0.1$ cm.
1. Two-tailed at 1%: $z_{0.005} = 2.576$. Boundaries: $25 \pm 2.576 \times 0.1 = 25 \pm 0.2576$. Critical region: $\bar{x} < 24.7424$ or $\bar{x} > 25.2576$.
2. Right-tailed at 5%: $z_{0.05} = 1.645$. Boundary: $25 + 1.645 \times 0.1 = 25.1645$. Critical region: $\bar{x} > 25.1645$.
3. $\bar{x} = 25.21$: in case 1, $25.21 < 25.2576$, so we do not reject H_0 at 1%; in case 2, $25.21 > 25.1645$, so we reject H_0 at 5%. Note how the conclusion depends on both the level and the type of test.

DID YOU KNOW? **Why 5%?**

The 5% convention was popularised by the statistician Ronald Fisher in the 1920s as a convenient, not sacred, threshold. In medicine, where a wrong conclusion can cost lives, 1% or even 0.1% levels are common; in preliminary agricultural field trials — for instance testing a new maize variety in the West Region — a 10% level may be tolerated so as not to miss a promising result.

KEY POINTS **Summary**

- $\alpha = P(\text{reject } H_0 \mid H_0 \text{ true})$, fixed in advance: 10%, 5% or 1%.

- Test statistic $Z = \dfrac{\bar{X} - \mu_0}{\sigma/\sqrt{n}} \sim N(0,1)$ under H_0 (use s instead of σ if $n \geq 30$ and σ unknown).
- Critical values: one tail 1.282/1.645/2.326; two tails 1.645/1.960/2.576.
- Two-tailed at 5% ⇒ 2.5% per tail, critical values ±1.96.
- Distinguish the critical value (Z-scale) from the boundary value of $\bar{X}$.
- Reject H_0 if z lies in the critical region, equivalently if the p-value $< \alpha$.
- p-value: two-tailed = 2× tail probability; one-tailed = tail probability in direction of H_1.

3 ERRORS IN HYPOTHESIS TESTING AND THE TESTING PROCEDURE

In the previous section we learned how to choose a **significance level** α, compute a **test statistic** and compare it with a **critical region**. But a hard question remains: since our decision is based on a sample, and samples vary by chance, can we ever be sure our decision is correct? The honest answer is **no**. Whatever we decide, there is a risk of being wrong. The whole art of hypothesis testing is to understand, measure and control these risks. That is the purpose of this section, which closes with the complete five-step procedure you will use in every GCE question.

3.1 The Two Possible Errors

An observation to start with. Suppose a magistrate in Bamenda must judge a defendant. The presumption of innocence plays the role of H_0: "the accused is innocent". The court examines the evidence (the sample) and decides. Whatever the verdict, two miscarriages of justice are possible: an innocent person may be convicted, or a guilty person may be acquitted. These two mistakes are not the same, and society considers convicting an innocent person the more serious one. Hypothesis testing works exactly the same way: rejecting a true H_0 is the "false alarm" we most want to control.

When we carry out a test, reality (which we cannot see) is in one of two states, and our decision is one of two actions. This gives four situations, summarised in the **truth table** below.

	H_0 is true	H_0 is false
Reject H_0	**Type I error** probability α	**Correct decision** probability $1-\beta$ (power)
Do not reject H_0	**Correct decision** probability $1-\alpha$	**Type II error** probability β

The four possible outcomes of a hypothesis test: decision (rows) against unknown reality (columns).

DEFINITION **Type I error**

A **Type I error** is committed when we reject H_0 although H_0 is actually true. By construction of the test, its probability is exactly the significance level:

$$P(\text{Type I error}) = P(\text{reject } H_0 \mid H_0 \text{ true}) = \alpha.$$

This is why α is chosen before looking at the data: the experimenter fixes in advance the greatest risk of a "false alarm" that he or she is willing to accept. At the 5% level, if H_0 is true, the sample will fall in the critical region purely by chance in about 5 cases out of 100.

DEFINITION **Type II error and power**

A **Type II error** is committed when we fail to reject H_0 although H_0 is actually false. Its probability is denoted β:

$$\beta = P(\text{do not reject } H_0 \mid H_0 \text{ false}).$$

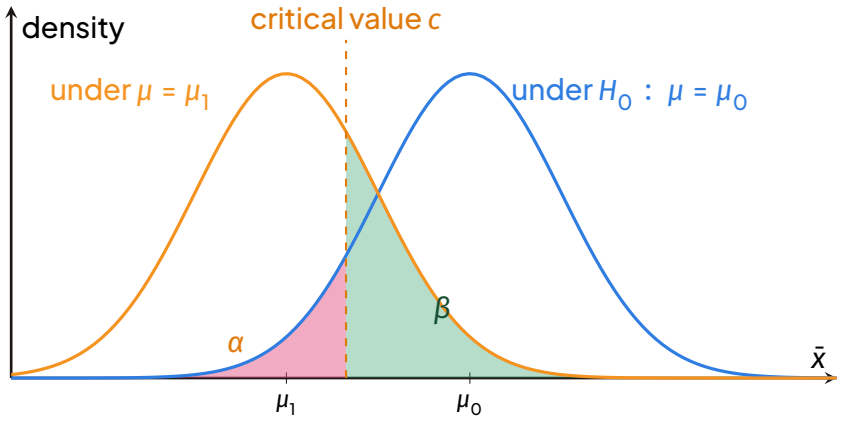
The quantity $1-\beta$ is called the **power** of the test: it is the probability of correctly detecting that H_0 is false.

WARNING **Why β has no single value**

Unlike α, the value of β is not fixed by the experimenter and is usually not a single number. "H_0 false" covers many possibilities (e.g. $\mu = 49$, $\mu = 45$, $\mu = 30$…), and β is different for each true value of the parameter. Computing β requires assuming a specific alternative value of the parameter.

3.2 Seeing α and β on the Same Picture

The best way to understand the two errors is to draw two sampling distributions on the same axes: the distribution of the sample mean $\bar{X}$ if H_0 is true (centred at μ_0), and its distribution if the true mean is some $\mu_1 < \mu_0$.



Left-tailed test. The pink area α is $P(\bar{X} < c \mid H_0 \text{ true})$; the green area β is $P(\bar{X} \geq c \mid \mu = \mu_1)$.

Read the figure carefully. The test rejects H_0 when $\bar{X} < c$. The pink area α is the chance of this happening when H_0 is true (false alarm). The green area β is the chance of $\bar{X}$ falling above c when the true mean is really μ_1 (missed detection). Everything about errors can be read from this picture.

PROPERTY **What β depends on**

For a left-tailed test as above (the same ideas hold for all tests):

1. Distance from μ_0. The further the true value μ_1 is from μ_0, the more the two curves separate, so the smaller β becomes (and the greater the power). It is easy to detect a big lie, hard to detect a small one.
2. The level α. If α is reduced (with n fixed), the critical value c moves towards μ_0, so the green area increases: β goes up and power goes down.
3. The sample size n. Since $\text{Var}(\bar{X}) = \sigma^2/n$, increasing n makes both curves narrower, so both α and β can be made small simultaneously.

WARNING **The α–β trade-off**

A very common GCE trap: "reducing α from 5% to 1% makes the test better in every way." False. With n fixed, decreasing α increases β: you make fewer false alarms but miss more

real effects. The only way to reduce both errors at once is to increase the sample size.

EXEMPLE **Computing β for a specific alternative**

In the figure above, suppose $\mu_0 = 48$, $\sigma/\sqrt{n} = 2$ and the left-tailed test at $\alpha = 0.05$ gives the critical value $c = 48 - 1.645 \times 2 = 44.71$. If the true mean is in fact $\mu_1 = 44$, then

$$\beta = P(\bar{X} \geq 44.71 \mid \mu = 44) = P\left(Z \geq \frac{44.71 - 44}{2}\right) = P(Z \geq 0.355) \approx 0.36.$$

So the power is $1 - \beta \approx 0.64$: this test detects a true mean of 44 only about 64 times out of 100. If instead the true mean were $\mu_1 = 42$, the same calculation gives $\beta = P(Z \geq 1.355) \approx 0.09$: a bigger deviation is much easier to detect.

DID YOU KNOW? **The power of a test**

The power $1-\beta$ is not examinable in detail at the GCE, but understanding it earns you marks in discussion questions. Medical researchers planning a drug trial routinely demand a power of at least 80%: they want at least an 80% chance of detecting the effect if it truly exists. This requirement is what tells them how many patients (n) to recruit.

3.3 The Five-Step Testing Procedure

We now assemble everything from the previous lessons into one standard procedure. Learn it as a checklist: GCE examiners award marks for each visible step.

METHOD **The five steps of a hypothesis test**

1. State the hypotheses. Write H_0 and H_1 clearly, with the parameter named (e.g. $H_0 : \mu = 50$ against $H_1 : \mu < 50$). The form of H_1 fixes whether the test is one-tailed or two-tailed.
2. Choose the significance level α and write down the corresponding critical region (using the sampling distribution under H_0 and the standard normal or t tables).
3. Compute the test statistic from the sample data, e.g. $z = \dfrac{\bar{x} - \mu_0}{\sigma/\sqrt{n}}$.
4. Decide: if the test statistic falls in the critical region, reject H_0; otherwise do not reject H_0.
5. Conclude in context: translate the statistical decision into a sentence about the orig-

inal problem, mentioning the level α.

WARNING **"Accept H_0" is misleading language**

When the test statistic is not in the critical region, we have not proved that H_0 is true — we have merely failed to find enough evidence against it (remember the court: "not guilty" does not mean "proved innocent"). Therefore always write "there is insufficient evidence, at the α level, to reject H_0" rather than "H_0 is true" or a bare "accept H_0".

METHOD **Writing a good conclusion sentence**

A full-mark conclusion has three ingredients:

1. the level: "at the 5% level of significance";
2. the decision: "there is (sufficient / insufficient) evidence to reject H_0";
3. the context: restate the claim in the words of the problem (the supplier, the bags, the mean mass…), never just symbols.

Model: "At the 5% level, there is sufficient evidence to conclude that the mean mass of the supplier's bags is less than 50 kg."

3.4 A Complete Worked Example: the Rice Bags of Ndop

WORKED EXAMPLE 5 **Testing the mean mass of bags of rice**

A supplier in Ndop claims that his bags of rice have a mean mass of 50 kg. Long experience shows the masses are normally distributed with standard deviation $\sigma = 2.4$ kg. A buyers' cooperative, suspecting the bags are underweight, weighs a random sample of 36 bags and obtains a mean mass of 49.1 kg. Test, at the 5% level of significance, whether the supplier's claim should be rejected. State the probability of a Type I error, and describe in words what a Type II error would mean here.

SOLUTION

Step 1 — Hypotheses. The cooperative suspects underweight bags, so this is a left-tailed test:

$$H_0 : \mu = 50 \qquad \text{against} \qquad H_1 : \mu < 50,$$

where μ is the true mean mass (in kg) of the supplier's bags.

Step 2 — Level and critical region. $\alpha = 0.05$, one-tailed. Under H_0,

$$\bar{X} \sim N\left(50, \tfrac{2.4^2}{36}\right), \qquad \text{so} \qquad Z = \frac{\bar{X} - 50}{2.4/\sqrt{36}} \sim N(0,1).$$

The critical value is $-z_{0.05} = -1.645$; the critical region is $z < -1.645$.
Step 3 — Test statistic.

$$z = \frac{49.1 - 50}{2.4/\sqrt{36}} = \frac{-0.9}{0.4} = -2.25.$$

Step 4 — Decision. Since $-2.25 < -1.645$, the test statistic lies in the critical region: we reject H_0.
Step 5 — Conclusion in context. At the 5% level of significance, there is sufficient evidence to conclude that the mean mass of the supplier's bags is less than 50 kg; the cooperative's suspicion is supported by the data.
Type I error. $P(\text{Type I error}) = \alpha = 0.05$: there is a 5% risk of accusing the supplier of underfilling when his bags actually average 50 kg.
Type II error (in words). A Type II error would occur if the bags really were underweight on average, but the sample happened to look acceptable, so we failed to reject the claim — the cooperative would keep buying short-weight bags. Its probability β would depend on the true mean: it is small if the true mean is far below 50 kg, and large if the true mean is only slightly below 50 kg.

KEY POINTS **Key points of this section**

- Type I error: reject a true H_0; its probability is α, chosen by the experimenter.
- Type II error: fail to reject a false H_0; its probability β depends on the true parameter value, on α and on n.
- Power $= 1 - \beta$: the probability of detecting a real effect.
- With n fixed, decreasing α increases β; only a larger sample reduces both.
- The five steps: hypotheses → level and critical region → test statistic → decision → conclusion in context.
- Never claim to have "proved" H_0: say "insufficient evidence to reject H_0 at the α level".

EXERCISE 3 **Errors and procedure**

A factory in Douala fills bottles supposed to contain $\mu_0 = 33$ cL of a soft drink, with known $\sigma = 1.5$ cL. An inspector tests $H_0 : \mu = 33$ against $H_1 : \mu \neq 33$ at the 5% level using a sample of $n = 25$ bottles.

1. Write down the critical region for the standardised statistic z.
2. The sample mean is $\bar{x} = 32.3$ cL. Carry out the five-step test and conclude in context.
3. State $P(\text{Type I error})$ and explain, in one sentence, what a Type II error would mean for the consumers.
4. Without calculation, explain what happens to β if the inspector switches to the 1% level with the same sample size.

SOLUTION **Exercise 1**

1. Two-tailed test at 5%: critical values ± 1.96; critical region $z < -1.96$ or $z > 1.96$.
2. Step 1: $H_0 : \mu = 33$, $H_1 : \mu \neq 33$. Step 2: $\alpha = 0.05$, critical region $|z| > 1.96$. Step 3: $z = \dfrac{32.3 - 33}{1.5/\sqrt{25}} = \dfrac{-0.7}{0.3} \approx -2.33$. Step 4: $-2.33 < -1.96$, so reject H_0. Step 5: at the 5% level, there is sufficient evidence that the mean content differs from 33 cL (in fact it appears to be lower).
3. $P(\text{Type I error}) = \alpha = 0.05$. A Type II error would mean the bottles really are under- (or over-) filled on average, yet the sample fails to reveal it, so consumers keep receiving incorrectly filled bottles.
4. Reducing α to 1% widens the acceptance region (critical values ± 2.576), so with n unchanged the probability β of a Type II error increases: the test becomes less able to detect a real deviation.

4 HYPOTHESIS TEST FOR A POPULATION MEAN

In the previous section we built the complete machinery of hypothesis testing: the null and alternative hypotheses H_0 and H_1, the significance level α, the test statistic, the critical region, and the two types of error. We now put this machinery to work on the most common problem of all: is the mean of a population equal to some claimed value μ_0? A bottling company claims its bottles contain 500 ml of juice; a school claims its candidates average 12 out of 20 in Mathematics. A sample is taken, the sample mean $\bar{x}$ is computed, and we must decide whether the gap between $\bar{x}$ and μ_0 is due to mere sampling fluctuation or to a real difference. This section answers that question rigorously.

4.1 Distribution of the Sample Mean

The whole test rests on one fundamental result, which you met in the chapter on sampling distributions.

PROPERTY **Distribution of the sample mean**

Let $X \sim N(\mu, \sigma^2)$ and let $X_1, X_2, \ldots, X_n$ be a random sample of size n from this population. Then the sample mean satisfies

$$\bar{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right), \qquad \text{so that} \qquad Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \sim N(0,1).$$

Moreover, by the **Central Limit Theorem**, if the population is not normal but n is large (in practice $n \geq 30$), then $\bar{X}$ is approximately $N(\mu, \sigma^2/n)$. The quantity $\sigma/\sqrt{n}$ is called the **standard error** of the mean. (The proof for the normal case follows from the fact that a linear combination of independent normal variables is normal, with $E(\bar{X}) = \mu$ and $\mathrm{Var}(\bar{X}) = \sigma^2/n$; it is examinable.)

The intuition is simple: individual values are scattered with spread σ, but averages are much more stable — the spread of $\bar{X}$ is $\sigma/\sqrt{n}$, which shrinks as n grows. So if $H_0 : \mu = \mu_0$ is true, we know exactly how $\bar{X}$ should behave, and we can measure how "surprising" an observed $\bar{x}$ is by standardising it.

4.2 The Three Cases: Choosing the Right Standard Error

In practice three situations arise, and the GCE examiners expect you to recognise each one.

Case 1: normal population, σ^2 known. This is the ideal case. If $X \sim N(\mu, \sigma^2)$ with σ^2 known, then under H_0 the statistic

$$Z = \frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}} \sim N(0,1) \quad \text{exactly.}$$

Case 2: σ^2 unknown, large sample ($n \geq 30$). We do not know σ^2, so we estimate it from the sample by the unbiased estimator $s^2 = \dfrac{\sum(x_i - \bar{x})^2}{n-1}$ and use

$$Z = \frac{\bar{X} - \mu_0}{s/\sqrt{n}} \approx N(0,1).$$

The approximation is excellent for large n because s is then a very stable estimate of σ.

Case 3: non-normal population, large sample. Even if the population distribution is skewed or unknown, the Central Limit Theorem guarantees that $\bar{X}$ is approximately normal for $n \geq 30$, so the same Z-statistic (with σ or s) remains valid.

WARNING **Use the unbiased estimate s^2**

When σ^2 is unknown, you must use the unbiased estimate

$$s^2 = \frac{\sum(x_i - \bar{x})^2}{n-1} = \frac{1}{n-1}\left(\sum x_i^2 - \frac{(\sum x_i)^2}{n}\right),$$

with divisor $n - 1$, not the population formula with divisor n. Using n underestimates the variance and makes the test too ready to reject H_0. Many marks are lost here.

WARNING **Divide by $\sqrt{n}$, not by n**

The standard error is $\frac{\sigma}{\sqrt{n}}$ (or $\frac{s}{\sqrt{n}}$). A very common error is to compute $Z = \frac{\bar{x} - \mu_0}{\sigma / n}$. Remember: $\mathrm{Var}(\bar{X}) = \frac{\sigma^2}{n}$, so its square root is $\frac{\sigma}{\sqrt{n}}$. Dividing by n instead of $\sqrt{n}$ inflates Z and leads to wrong conclusions.

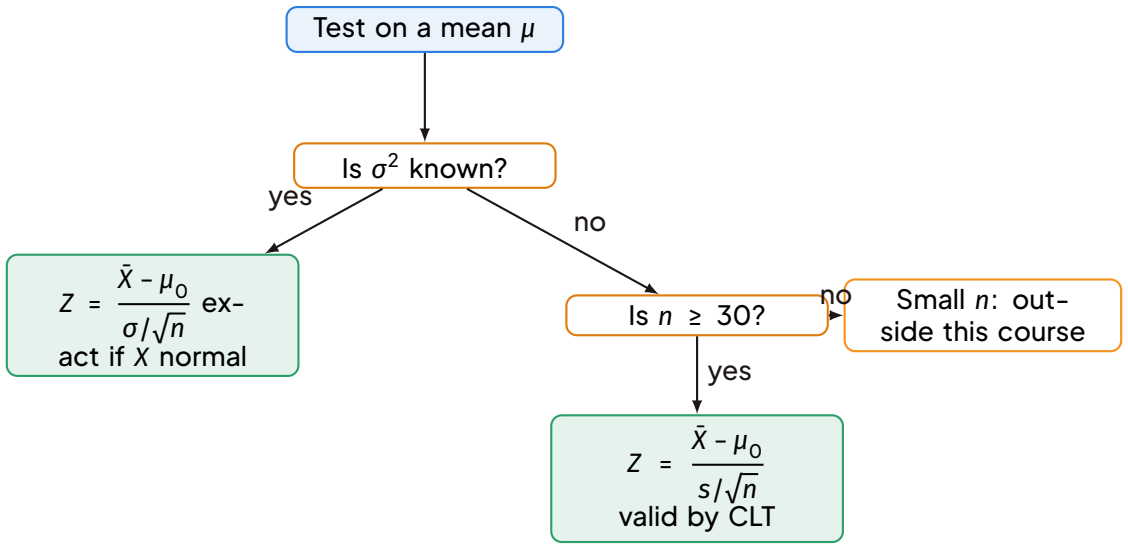
4.3 The Complete Z-Test Procedure

METHOD **Z-test for a population mean**

1. State the hypotheses. $H_0 : \mu = \mu_0$ against $H_1 : \mu \neq \mu_0$ (two-tailed), or $H_1 : \mu > \mu_0$ or $H_1 : \mu < \mu_0$ (one-tailed), choosing H_1 from the wording of the problem.
2. Choose the significance level α (usually 5% or 1%) and read the critical value(s): 1.645 (one-tailed, 5%), 1.96 (two-tailed, 5%), 2.326 (one-tailed, 1%), 2.576 (two-tailed, 1%).
3. Compute the test statistic.

$$z = \frac{\bar{x} - \mu_0}{\sigma / \sqrt{n}} \quad (\sigma \text{ known}) \qquad \text{or} \qquad z = \frac{\bar{x} - \mu_0}{s / \sqrt{n}} \quad (\sigma \text{ unknown, } n \geq 30).$$

4. Decide. Reject H_0 if z falls in the critical region: $|z| > 1.96$ (two-tailed, 5%), $z > 1.645$ or $z < -1.645$ (one-tailed, 5%).
5. Conclude in context, in plain language, stating whether the evidence against H_0 is significant at the level α.



Flowchart for choosing the correct test statistic for a population mean.

4.4 Worked Example 1: Two-Tailed Test, σ Known

WORKED EXAMPLE 6 **The juice bottling machine**

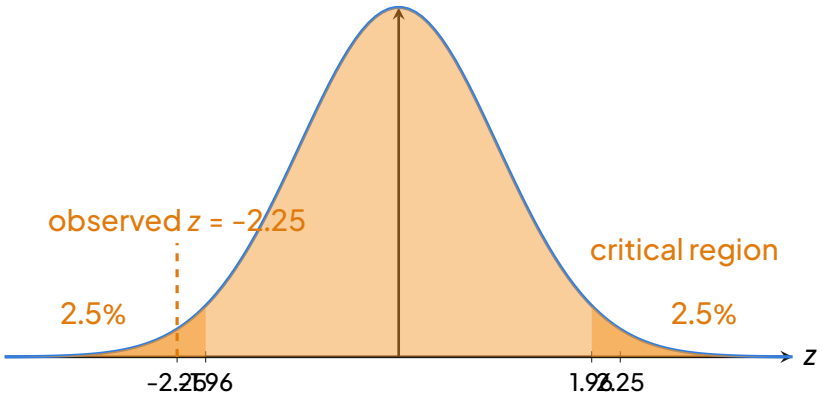
A machine in a Douala factory fills bottles so that the volume dispensed is $X \sim N(\mu, \sigma^2)$ with $\sigma = 4$ ml known from long experience. The machine is supposed to deliver $\mu = 500$ ml. After maintenance, a random sample of 25 bottles gives a mean volume of $\bar{x} = 498.2$ ml. Test at the 5% level whether the mean volume has changed.

SOLUTION

Solution. Step 1. $H_0 : \mu = 500$; $H_1 : \mu \neq 500$ ("changed" means two-tailed).
Step 2. Two-tailed test at $\alpha = 5\%$: critical values ± 1.96. Reject H_0 if $|z| > 1.96$.
Step 3. σ is known, X normal, so

$$z = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}} = \frac{498.2 - 500}{4/\sqrt{25}} = \frac{-1.8}{0.8} = -2.25.$$

Step 4. $|z| = 2.25 > 1.96$: the observed value lies in the critical region. We reject H_0.
Step 5. Conclusion: at the 5% level of significance there is sufficient evidence that the mean volume is no longer 500 ml; the sample suggests the machine is under-filling, so it should be readjusted.



Two-tailed test at 5%: the observed value $z = -2.25$ falls inside the left critical region, so H_0 is rejected.

4.5 Worked Example 2: One-Tailed Test, σ Unknown, Large Sample

WORKED EXAMPLE 7 **Average score in a school**

The principal of a large college in Yaoundé claims that the mean score of his Upper Sixth students in a national mock examination is more than 12 out of 20. A random sample of 64 scripts gives $\bar{x} = 12.5$ and $\sum(x_i - \bar{x})^2 = 567$. Test the principal's claim at the 5% level of significance.

SOLUTION

Solution. Step 1. The claim is "more than 12", so it goes into H_1:

$$H_0 : \mu = 12 \qquad H_1 : \mu > 12 \quad \text{(one-tailed, right)}.$$

Step 2. One-tailed test at $\alpha = 5\%$: reject H_0 if $z > 1.645$.
Step 3. σ^2 is unknown but $n = 64 \geq 30$, so we use the unbiased estimate:

$$s^2 = \frac{\sum(x_i - \bar{x})^2}{n - 1} = \frac{567}{63} = 9, \qquad s = 3.$$

By the Central Limit Theorem,

$$z = \frac{\bar{x} - \mu_0}{s/\sqrt{n}} = \frac{12.5 - 12}{3/\sqrt{64}} = \frac{0.5}{0.375} = 1.333.$$

Step 4. $z = 1.333 < 1.645$: not in the critical region. We do not reject H_0.
Step 5. Conclusion: at the 5% level there is insufficient evidence to support the principal's claim that the mean score exceeds 12. Note carefully: we have not proved that $\mu = 12$; we have simply found that the data do not provide strong enough evidence against it.

4.6 Link with Confidence Intervals (GCE Extra)

There is a beautiful and examinable connection between a two-tailed test and a confidence interval. The 95% confidence interval for μ is

$$\bar{x} - 1.96\,\frac{\sigma}{\sqrt{n}} \le \mu \le \bar{x} + 1.96\,\frac{\sigma}{\sqrt{n}}.$$

Now the two-tailed test at 5% accepts $H_0 : \mu = \mu_0$ exactly when

$$\left|\frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}}\right| \le 1.96 \iff \bar{x} - 1.96\frac{\sigma}{\sqrt{n}} \le \mu_0 \le \bar{x} + 1.96\frac{\sigma}{\sqrt{n}},$$

that is, exactly when μ_0 lies inside the 95% confidence interval.

PROPERTY **Duality between tests and confidence intervals**

A two-tailed test of $H_0 : \mu = \mu_0$ at level α rejects H_0 if and only if μ_0 lies outside the $(1 - \alpha)$ confidence interval for μ. (Not required to be proved at GCE, but frequently examined as a checking technique.)

METHOD **Checking a conclusion with a confidence interval**

1. Compute the 95% confidence interval $\left[\bar{x} - 1.96\,\frac{s}{\sqrt{n}},\ \bar{x} + 1.96\,\frac{s}{\sqrt{n}}\right]$.
2. If μ_0 lies outside the interval, the two-tailed test at 5% rejects H_0; if μ_0 lies inside, H_0 is not rejected.
3. Use this as a quick cross-check of your z-calculation.

EXEMPLE **Checking the bottling example**

For the juice machine, the 95% confidence interval is

$$498.2 \pm 1.96 \times \frac{4}{\sqrt{25}} = 498.2 \pm 1.568 = [496.63\,;\ 499.77]\ \text{ml}.$$

Since $\mu_0 = 500$ lies outside this interval, the two-tailed test at 5% rejects H_0 — exactly as we found with $z = -2.25$.

4.7 Interpreting the Conclusion for a Non-Specialist

A statistical conclusion must be communicated honestly. "Reject H_0 at the 5% level" means: if the claimed mean were true, a sample mean as extreme as the one observed would occur in

fewer than 5% of samples — so the claim is not credible. It does **not** mean the claim is certainly false: there is still a probability α (a Type I error) of rejecting a true H_0. Conversely, "do not reject H_0" does **not** prove H_0 true; it means the evidence is insufficient. Always finish with one clear sentence in the language of the problem, mentioning the significance level, as we did in both worked examples.

KEY POINTS **Key points**

- Under $H_0 : \mu = \mu_0$: $\bar{X} \sim N(\mu_0, \sigma^2/n)$; the test statistic is $z = \dfrac{\bar{x} - \mu_0}{\sigma/\sqrt{n}}$.
- σ^2 unknown and $n \geq 30$: replace σ by $s = \sqrt{\dfrac{\sum(x_i - \bar{x})^2}{n-1}}$; the CLT validates the Z-test even for non-normal populations.
- Standard error $= \sigma/\sqrt{n}$, never σ/n.
- Two-tailed 5%: reject if $|z| > 1.96$; one-tailed 5%: reject if $z > 1.645$ (right) or $z < -1.645$ (left).
- Two-tailed test at 5% $\iff$ check whether μ_0 is outside the 95% confidence interval.

EXERCISE 4 **Practice**

A factory in Bafoussam produces sugar packets with masses normally distributed and $\sigma = 5$ g. The nominal mean mass is 1000 g. A random sample of 36 packets has mean 997.5 g.

1. Test at the 5% level whether the mean mass differs from 1000 g.
2. Test at the 1% level whether the mean mass is less than 1000 g.
3. Compute the 95% confidence interval for μ and confirm your answer to part 1.

SOLUTION **Exercise 1**

1. $H_0 : \mu = 1000$, $H_1 : \mu \neq 1000$. $z = \dfrac{997.5 - 1000}{5/\sqrt{36}} = \dfrac{-2.5}{0.8333} = -3.0$. Since $|-3| > 1.96$, reject H_0: the mean mass has changed.
2. $H_1 : \mu < 1000$, one-tailed at 1%: critical value -2.326. Since $z = -3.0 < -2.326$, reject H_0: significant evidence at the 1% level that the packets are underweight.
3. $997.5 \pm 1.96 \times \dfrac{5}{6} = 997.5 \pm 1.633$, giving $[995.87\,;\,999.13]$ g. Since $1000 \notin [995.87; 999.13]$, the two-tailed test at 5% rejects H_0, confirming part 1.

5 TEST FOR THE DIFFERENCE BETWEEN TWO MEANS (LARGE SAMPLES)

In the previous section we learnt how to test a claim about one population mean: is the mean mass of bags of cement really 50 kg? In real life, however, the most interesting questions are comparative. Is a new maize variety more productive than the traditional one? Do students of School A perform better at the GCE than students of School B? Does a new drug reduce blood pressure more than the old one? In all these questions we have **two populations**, and we want to compare their means μ_1 and μ_2. This section extends the machinery of hypothesis testing to exactly that situation, in the case where both samples are large.

5.1 From One Mean to the Difference of Two Means

The comparative question.

Suppose an agronomist in the West Region wants to know whether variety 1 of maize yields more per hectare than variety 2. She cannot measure the whole populations, so she takes a sample of n_1 fields planted with variety 1 and an independent sample of n_2 fields planted with variety 2, obtaining sample means $\bar{x}_1$ and $\bar{x}_2$. Suppose she finds $\bar{x}_1 = 3.2$ tonnes and $\bar{x}_2 = 2.9$ tonnes. Does this prove that $\mu_1 > \mu_2$? Not necessarily: even if the two varieties were equally productive, sampling fluctuation alone could produce $\bar{x}_1 > \bar{x}_2$. The question is whether the observed difference $\bar{x}_1 - \bar{x}_2$ is too large to be explained by chance. This is exactly the spirit of hypothesis testing, applied to the parameter $\mu_1 - \mu_2$.

DEFINITION **Hypotheses for the comparison of two means**

Let μ_1 and μ_2 be the means of two populations. A test comparing them uses a null hypothesis of the form

$$H_0 : \mu_1 - \mu_2 = d \qquad \text{(most often } d = 0\text{, i.e. } H_0 : \mu_1 = \mu_2\text{)},$$

against one of the alternatives:

- $H_1 : \mu_1 - \mu_2 \neq d$ (two-tailed test);
- $H_1 : \mu_1 - \mu_2 > d$ (one-tailed, right tail);
- $H_1 : \mu_1 - \mu_2 < d$ (one-tailed, left tail).

METHOD **Stating H_0 with $d \neq 0$**

When the claim is not "the means are equal" but "mean 1 exceeds mean 2 by at least d", proceed as follows:

1. Write the boundary of the claim as the null hypothesis: $H_0 : \mu_1 - \mu_2 = d$.

2. Write the alternative according to what must be demonstrated: to prove "exceeds by more than d", use $H_1 : \mu_1 - \mu_2 > d$ (right-tailed).
3. Use the same test statistic as for $d = 0$, subtracting d in the numerator:

$$Z = \frac{\bar{x}_1 - \bar{x}_2 - d}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}.$$

For example, "the new variety yields at least 0.5 tonne per hectare more" gives $H_0 : \mu_1 - \mu_2 = 0.5$ against $H_1 : \mu_1 - \mu_2 > 0.5$.

5.2 Distribution of the Difference of Two Sample Means

To build a test we need the sampling distribution of the statistic $\bar{X}_1 - \bar{X}_2$ under H_0. Recall two facts from earlier chapters: a linear combination of independent normal variables is normal; means add, and variances of independent variables add.

PROPERTY **Distribution of the difference of two independent sample means**

Let X_1 and X_2 be two independent random variables with

$$\bar{X}_1 \sim N\left(\mu_1, \tfrac{\sigma_1^2}{n_1}\right), \qquad \bar{X}_2 \sim N\left(\mu_2, \tfrac{\sigma_2^2}{n_2}\right),$$

where $\bar{X}_1$ and $\bar{X}_2$ are the means of independent samples of sizes n_1 and n_2. Then

$$\bar{X}_1 - \bar{X}_2 \sim N\left(\mu_1 - \mu_2, \frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}\right).$$

Justification (instructive, not required at the GCE). The mean of a difference is the difference of the means: $E(\bar{X}_1 - \bar{X}_2) = \mu_1 - \mu_2$. For independent variables, $\operatorname{Var}(\bar{X}_1 - \bar{X}_2) = \operatorname{Var}(\bar{X}_1) + \operatorname{Var}(\bar{X}_2) = \frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}$, because $\operatorname{Var}(-\bar{X}_2) = (-1)^2 \operatorname{Var}(\bar{X}_2)$. A sum of independent normals is normal, which completes the result. When the populations are not normal but n_1, n_2 are large, the Central Limit Theorem makes the result approximately true.

WARNING **Variances add — even for a difference!**

The most frequent error at the GCE is to subtract the variances. Even though we study a

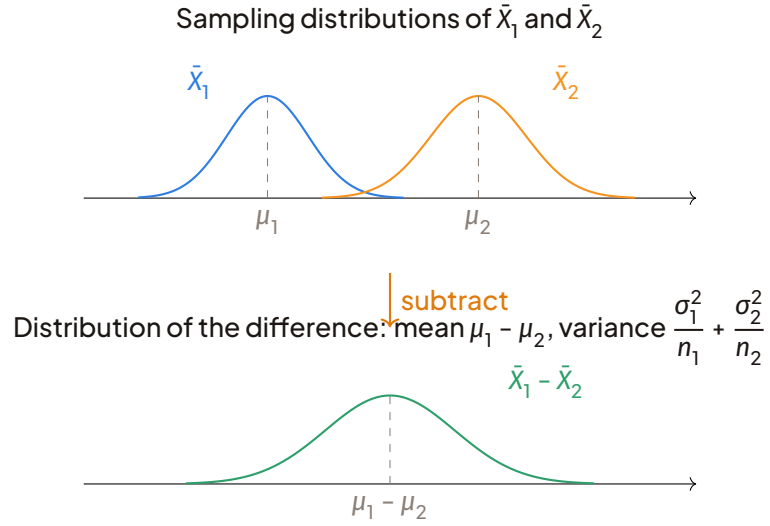
difference of means, the variance of the difference is the sum

$$\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}.$$

Reason: $\mathrm{Var}(-\bar{X}_2) = (-1)^2\,\mathrm{Var}(\bar{X}_2) = +\mathrm{Var}(\bar{X}_2)$. Subtracting the variances would give a standard error that is too small and could even be negative. Never write $\frac{\sigma_1^2}{n_1} - \frac{\sigma_2^2}{n_2}$.

WARNING **The two samples must be independent**

This test is valid only if the two samples are drawn **independently**: the choice of individuals in sample 1 must not influence sample 2. It does not apply to paired data — for example the same patients measured before and after a treatment, or the same students tested in June and in September. Paired data are analysed by reducing the pairs to a single sample of differences, which is a one-sample problem.



Two independent sampling distributions (top) combine into the distribution of $\bar{X}_1 - \bar{X}_2$ (bottom): the means subtract, the variances add.

5.3 The Test Statistic and Conditions of Validity

Standardising the difference under $H_0 : \mu_1 - \mu_2 = d$ gives the test statistic.

KEY POINTS **Test statistic for two means, large samples**

Under $H_0 : \mu_1 - \mu_2 = d$,

$$Z = \frac{\bar{X}_1 - \bar{X}_2 - d}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}} \sim N(0,1).$$

When σ_1^2 and σ_2^2 are unknown (the usual case), they are replaced by the sample estimates s_1^2 and s_2^2; this is legitimate because n_1 and n_2 are large.

Conditions of validity.

- The two samples are **independent** random samples.
- Both samples are **large** (in practice $n_1 \geq 30$ and $n_2 \geq 30$), so that the Central Limit Theorem applies and s_1^2, s_2^2 are reliable estimates of σ_1^2, σ_2^2.

METHOD **Z-test for the difference between two means (large samples)**

1. State the hypotheses: $H_0 : \mu_1 - \mu_2 = d$ against the appropriate H_1 (one- or two-tailed), and choose the significance level α.
2. Compute the test statistic
$$z = \frac{\bar{x}_1 - \bar{x}_2 - d}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}.$$
3. Determine the critical value(s) from the standard normal table: e.g. 1.645 (one tail, 5%), 1.96 (two tails, 5%), 2.576 (two tails, 1%).
4. Compare: reject H_0 if z falls in the critical region ($z > z_\alpha$, $z < -z_\alpha$, or $|z| > z_{\alpha/2}$).
5. Conclude in context: state clearly what the decision means for the original problem.

Step 1 — Hypotheses: $H_0 : \mu_1 - \mu_2 = d$; choose H_1 (one/two-tailed) and α.

Step 2 — Statistic: $z = \dfrac{\bar{x}_1 - \bar{x}_2 - d}{\sqrt{s_1^2/n_1 + s_2^2/n_2}}$ (variances add).

Step 3 — Critical value: read z_α or $z_{\alpha/2}$ from the $N(0,1)$ table.

Step 4 — Decision: reject H_0 if z lies in the critical region.

Step 5 — Conclusion: interpret in the context of the problem (yields, scores, …).

The five steps of a two-sample Z-test, to be reproduced in every worked solution.

5.4 Worked Examples

WORKED EXAMPLE 8 **Comparing two maize varieties in the West Region**

An agricultural station near Bafoussam compares two maize varieties. Variety 1 is grown on $n_1 = 60$ independent plots, giving a mean yield of $\bar{x}_1 = 3.2$ tonnes per hectare with variance $s_1^2 = 0.64$. Variety 2 is grown on $n_2 = 50$ plots, giving $\bar{x}_2 = 2.9$ tonnes per hectare with variance $s_2^2 = 0.50$. Test, at the 5% level, whether variety 1 gives a higher mean yield than variety 2.

SOLUTION

Step 1 — Hypotheses. Let μ_1, μ_2 be the true mean yields.

$$H_0 : \mu_1 - \mu_2 = 0 \quad \text{against} \quad H_1 : \mu_1 - \mu_2 > 0 \quad \text{(right-tailed)}, \qquad \alpha = 0.05.$$

Step 2 — Test statistic. The samples are independent and large, so

$$z = \frac{\bar{x}_1 - \bar{x}_2 - 0}{\sqrt{\dfrac{s_1^2}{n_1} + \dfrac{s_2^2}{n_2}}} = \frac{3.2 - 2.9}{\sqrt{\dfrac{0.64}{60} + \dfrac{0.50}{50}}} = \frac{0.3}{\sqrt{0.010667 + 0.010}} = \frac{0.3}{\sqrt{0.020667}} = \frac{0.3}{0.14376} \approx 2.09.$$

Step 3 — Critical value. One-tailed at 5%: $z_{0.05} = 1.645$. Critical region: $z > 1.645$. Step 4 — Decision. Since $2.09 > 1.645$, we reject H_0. Step 5 — Conclusion. At the 5% level there is significant evidence that variety 1 has a higher mean yield than variety 2. The station can recommend variety 1 to farmers of the West Region, while noting that the observed advantage is about 0.3 tonne per hectare.

WORKED EXAMPLE 9 **Comparing mean GCE scores of two schools**

A random sample of $n_1 = 80$ candidates from School A has a mean score of $\bar{x}_1 = 12.4$ with variance $s_1^2 = 16.0$; an independent random sample of $n_2 = 100$ candidates from School B has mean $\bar{x}_2 = 11.9$ with variance $s_2^2 = 20.0$. Test at the 1% level whether there is any difference between the mean scores of the two schools.

SOLUTION

Step 1 — Hypotheses. "Any difference" means a two-tailed test:

$$H_0 : \mu_1 - \mu_2 = 0 \quad \text{against} \quad H_1 : \mu_1 - \mu_2 \neq 0, \qquad \alpha = 0.01.$$

Step 2 — Test statistic.

$$z = \frac{12.4 - 11.9}{\sqrt{\dfrac{16.0}{80} + \dfrac{20.0}{100}}} = \frac{0.5}{\sqrt{0.20 + 0.20}} = \frac{0.5}{\sqrt{0.40}} = \frac{0.5}{0.63246} \approx 0.79.$$

Step 3 — Critical values. Two-tailed at 1%: $z_{0.005} = 2.576$. Critical region: $|z| > 2.576$. Step 4 — Decision. Since $|0.79| < 2.576$, we do not reject H_0. Step 5 — Conclusion. At the 1% level there is no significant evidence of a difference between the mean scores of the two schools.

5.5 Interpreting a Non-Significant Difference

The last example illustrates a subtle but essential point. We did not prove that $\mu_1 = \mu_2$; we only failed to find evidence against that claim. The observed difference 0.5 might reflect a real but small gap that our samples were not large enough to detect.

WARNING **"Not significant" does not mean "equal"**

When H_0 is not rejected, the correct conclusion is: "the data do not provide sufficient evidence, at the α level, of a difference between the means." It is a mistake to write "the two means are equal." A non-significant result may arise because the true difference is small, because the samples are too small, or because the variability is large. This is why we say we fail to reject H_0 rather than we accept it.

DID YOU KNOW? **Why this test matters in Cameroon**

Comparative tests of two means are the everyday tool of agricultural research stations (comparing crop varieties), of the Ministry of Public Health (comparing average recovery times under two treatments) and of education analysts (comparing performance between schools or between years). Whenever you read "on average, group A does better than

group B", ask yourself: was the difference tested, or could it be due to sampling fluctuation alone?

EXERCISE 5 **Filling stations**

Two filling stations in Douala are suspected of delivering different average volumes of fuel per "litre" sold. Independent measurements on $n_1 = 40$ pumps of station 1 give $\bar{x}_1 = 0.98$ L with $s_1^2 = 0.0064$; on $n_2 = 36$ pumps of station 2, $\bar{x}_2 = 1.01$ L with $s_2^2 = 0.0081$. Test at the 5% level whether there is a difference between the two stations.

SOLUTION **Exercise 1**

$H_0 : \mu_1 - \mu_2 = 0$ against $H_1 : \mu_1 - \mu_2 \neq 0$ (two-tailed), $\alpha = 0.05$.

$$z = \frac{0.98 - 1.01}{\sqrt{\dfrac{0.0064}{40} + \dfrac{0.0081}{36}}} = \frac{-0.03}{\sqrt{0.00016 + 0.000225}} = \frac{-0.03}{\sqrt{0.000385}} = \frac{-0.03}{0.019621} \approx -1.53.$$

Critical values: ± 1.96. Since $|-1.53| < 1.96$, we do not reject H_0: at the 5% level the data do not show a significant difference between the mean volumes delivered by the two stations. Note carefully: we have not proved the stations deliver equal volumes; the evidence is simply insufficient to conclude that they differ.

KEY POINTS **Key points of the section**

- To compare two means, test $H_0 : \mu_1 - \mu_2 = d$ (usually $d = 0$) against a one- or two-tailed alternative.
- For independent samples: $\bar{X}_1 - \bar{X}_2 \sim N\left(\mu_1 - \mu_2, \dfrac{\sigma_1^2}{n_1} + \dfrac{\sigma_2^2}{n_2}\right)$ — means subtract, variances add.
- Test statistic: $z = \dfrac{\bar{x}_1 - \bar{x}_2 - d}{\sqrt{s_1^2/n_1 + s_2^2/n_2}}$, valid for large independent samples.
- Follow the five steps: hypotheses, statistic, critical value, decision, conclusion in context.
- Failing to reject H_0 never proves that the two means are equal.

6 APPLICATIONS, CONSOLIDATION AND EXAM PRACTICE

In the previous section you learnt how to compare two population means using large independent samples. You now possess the complete toolkit of hypothesis testing: hypotheses, significance levels, critical regions, errors, the five-step procedure, and tests on one mean or on the difference of two means. This final section shows you where these tools are actually used in Cameroon and beyond, how to choose the right test from an exam question, and how to present your work so that you collect every method mark on the GCE paper.

6.1 Applications in Quality Control

Manufacturers must guarantee that the mean content of a package matches what is printed on the label. A company in Douala filling 50 kg bags of cement, or a factory bottling 50 cl of fruit juice, cannot weigh every single item; instead, a random sample is taken and a hypothesis test decides whether the machine needs adjustment.

EXEMPLE **Checking the mean content of sugar packets**

A machine fills packets labelled 1 kg of sugar. Past experience gives $\sigma = 15$ g. A sample of 60 packets has mean mass $\bar{x} = 995$ g. Has the machine drifted? Here the natural test is **two-tailed** ($H_0 : \mu = 1000$, $H_1 : \mu \neq 1000$) because both under-filling (cheating customers) and over-filling (losing money) matter. The test statistic is

$$z = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}} = \frac{995 - 1000}{15/\sqrt{60}} = \frac{-5}{1.9365} \approx -2.58.$$

At the 5% level the critical values are ± 1.96; since $-2.58 < -1.96$, we reject H_0: there is evidence, at the 5% level, that the mean content differs from 1 kg.

Notice how the context dictates the type of test: if a consumer association only suspects under-filling, a one-tailed test ($H_1 : \mu < 1000$) is appropriate.

6.2 Applications in Health

Doctors compare treatments by testing the difference of two means: mean recovery time under a new drug versus the standard one, or mean systolic blood pressure in two groups of patients.

WORKED EXAMPLE 10 **Comparing two treatments for malaria**

A hospital in Yaoundé compares recovery times (in days). Treatment A (standard): $n_1 = 80$, $\bar{x}_1 = 5.2$, $s_1 = 1.4$. Treatment B (new): $n_2 = 100$, $\bar{x}_2 = 4.7$, $s_2 = 1.6$. Test at the 1% level whether the new treatment reduces the mean recovery time.

SOLUTION

Step 1 — Hypotheses. $H_0 : \mu_1 = \mu_2$ (no difference); $H_1 : \mu_1 > \mu_2$ (the standard treatment is slower, i.e. the new treatment is faster). One-tailed test.
Step 2 — Level. $\alpha = 1\%$; critical value $z = 2.326$ (right tail).
Step 3 — Statistic. Large samples, so we use the sample variances in place of the population variances:

$$z = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\dfrac{s_1^2}{n_1} + \dfrac{s_2^2}{n_2}}} = \frac{5.2 - 4.7}{\sqrt{\dfrac{1.4^2}{80} + \dfrac{1.6^2}{100}}} = \frac{0.5}{\sqrt{0.0245 + 0.0256}} = \frac{0.5}{0.2238} \approx 2.23.$$

Step 4 — Decision. $2.23 < 2.326$: the statistic is not in the critical region, so we do not reject H_0.
Step 5 — Conclusion in context. At the 1% level there is insufficient evidence that treatment B reduces the mean recovery time. (Note: at the 5% level the conclusion would change, since $2.23 > 1.645$ — the choice of α matters!)

6.3 Applications in Education and Agriculture

The same machinery answers questions such as: "Do students taught with the new method score higher on average?" or "Does the improved maize variety give a higher mean yield per hectare than the local variety?" In each case, two large independent samples (two schools, two groups of farms in the West Region) are compared with the two-sample z-statistic, and the mean incomes of two groups of traders can be compared in exactly the same way.

EXEMPLE **Mean marks of two schools**

School X: $n_1 = 50$, $\bar{x}_1 = 12.4$, $s_1 = 3.1$. School Y: $n_2 = 60$, $\bar{x}_2 = 11.6$, $s_2 = 3.4$. Test at the 5% level whether the mean marks differ (two-tailed):

$$z = \frac{12.4 - 11.6}{\sqrt{\dfrac{3.1^2}{50} + \dfrac{3.4^2}{60}}} = \frac{0.8}{\sqrt{0.1922 + 0.1927}} = \frac{0.8}{0.6204} \approx 1.29.$$

Since $|1.29| < 1.96$, we do not reject H_0: the observed difference in mean marks is not significant at the 5% level.

6.4 Choosing the Significance Level Sensibly

The level α is the probability of a Type I error (rejecting a true H_0). Its choice should reflect the cost of each error in context.

- If a Type I error is **very costly** — e.g. declaring an effective drug ineffective, or stopping a whole production line unnecessarily — choose a **small** α such as 1%.

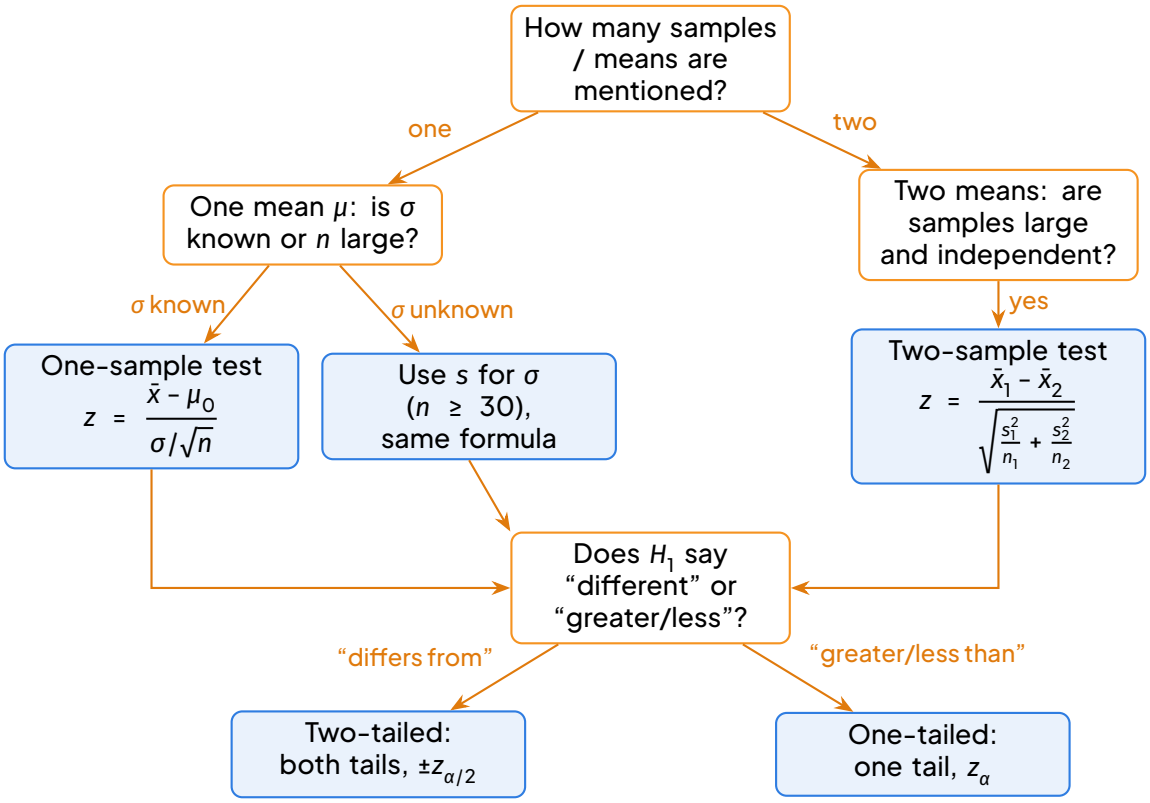
- If a Type II error (accepting a false H_0) is **more dangerous** — e.g. failing to detect under-filled medicine bottles, or missing a real improvement in patient survival — a larger α such as 5% or 10% makes rejection easier and so reduces β.

WARNING **Sample size and the two errors**

For a fixed sample size, decreasing α increases β and vice versa — you cannot shrink both at once. But **increasing the sample size** reduces the standard error, which makes the test sharper: a redesigned test with a larger n can achieve a smaller α and a smaller β simultaneously. Always mention this when a question asks you to "comment" or "suggest an improvement".

6.5 Choosing the Correct Test: A Decision Tree

Exam questions rarely say "use a two-sample test". You must read the wording and decide. The figure below summarises the reasoning.



Decision tree for selecting the correct test from a problem statement.

METHOD **Justifying one-tailed versus two-tailed in one sentence**

Look at the claim in the question, then write one sentence, for example:

- "The test is two-tailed because the question asks whether the mean differs from 50 cl, with no direction specified."
- "The test is one-tailed (left tail) because the consumer association claims the mean content is less than 50 cl."

The key words are: different / changed / not equal ⇒ two-tailed; greater / increased / more or less / reduced / lower ⇒ one-tailed.

6.6 Summary of All Test Statistics

PROPERTY **Summary table of tests in this chapter**

Not examinable as a proof, but every formula must be known by heart.

Situation	Hypotheses	Test statistic
One mean, σ known	$H_0 : \mu = \mu_0$	$z = \dfrac{\bar{x} - \mu_0}{\sigma/\sqrt{n}}$
One mean, σ unknown, n large	$H_0 : \mu = \mu_0$	$z = \dfrac{\bar{x} - \mu_0}{s/\sqrt{n}}$
Difference of two means, large independent samples	$H_0 : \mu_1 = \mu_2$	$z = \dfrac{\bar{x}_1 - \bar{x}_2}{\sqrt{\dfrac{s_1^2}{n_1} + \dfrac{s_2^2}{n_2}}}$

Critical values: two-tailed 5%: ±1.96; 1%: ±2.576. One-tailed 5%: 1.645; 1%: 2.326.

6.7 GCE Exam Technique: Earning Every Method Mark

GCE mark schemes award marks step by step. Even if your final arithmetic is wrong, correctly stated hypotheses, a correct standard error and a correct critical region each carry marks.

METHOD **Checklist for a complete exam answer**

1. State hypotheses using symbols and define them: $H_0 : \mu = 50$, $H_1 : \mu < 50$, where μ is the mean content in cl.
2. State α and the critical region: e.g. "5% one-tailed, reject H_0 if $z < -1.645$".
3. Compute the test statistic, showing the standard error explicitly: $\sigma/\sqrt{n}$ or

$\sqrt{s_1^2/n_1 + s_2^2/n_2}$.

4. Compare and decide: "$z = -2.10 < -1.645$, so z lies in the critical region; reject H_0."
5. Conclude in context: "There is evidence, at the 5% level, that the mean content is less than 50 cl."

WARNING **The conclusion must mention the context**

Writing only "reject H_0" or "$\mu < 50$" loses the final mark. The examiner wants a sentence about the real situation: the sugar packets, the recovery times, the maize yields. Symbols alone are never a conclusion.

The figure below shows a model answer presented as the examiner expects it, with the mark allocation indicated.

Question. A bottling machine should deliver a mean of 50 cl. A sample of 64 bottles gives $\bar{x} = 49.6$ cl, $s = 1.2$ cl. Test at 5% whether the mean content is less than 50 cl.

$H_0 : \mu = 50$; $H_1 : \mu < 50$ — M1
One-tailed, $\alpha = 5\%$: reject H_0 if $z < -1.645$ — B1
$z = \dfrac{49.6 - 50}{1.2/\sqrt{64}} = \dfrac{-0.4}{0.15} = -2.67$ — M1 A1
$-2.67 < -1.645 \Rightarrow$ reject H_0 — M1
At the 5% level, the mean content is less than 50 cl. — A1

6 marks: hypotheses, region, method, value, decision, context

Model answer of a full GCE question with mark allocation (M = method, B = independent accuracy, A = accuracy).

6.8 Remediation Checkpoints: Common Errors

WARNING **Frequent mistakes to eliminate**

- Wrong critical value: using 1.96 for a one-tailed 5% test (correct: 1.645) or 1.645 for a two-tailed test (correct: ±1.96). Always match the value to the type of test.
- Wrong standard error: forgetting $\sqrt{n}$, writing σ/n, or for two samples adding the variances without dividing by the sample sizes. Write the denominator out in full before substituting.
- Sign confusion: a negative z in a right-tailed test is automatically non-significant — check the direction of H_1 before concluding.

- Conclusion without context, or claiming H_0 is "proved". We only ever say the evidence is sufficient or insufficient at the stated level.
- Mixing samples: in two-sample problems, keep $\bar{x}_1, s_1, n_1$ and $\bar{x}_2, s_2, n_2$ clearly labelled.

WORKED EXAMPLE 11 **Mixed revision: identify and carry out the test**

A farmers' cooperative claims that a new fertiliser increases the mean yield of cassava, previously 12 tonnes per hectare. A sample of 49 farms using the fertiliser gives $\bar{x} = 12.7$ tonnes and $s = 2.1$ tonnes. Test the claim at the 1% level.

SOLUTION

Wording "increases" ⇒ one mean, one-tailed (right tail). $H_0 : \mu = 12$; $H_1 : \mu > 12$. Critical value at 1%: $z = 2.326$.

$$z = \frac{12.7 - 12}{2.1/\sqrt{49}} = \frac{0.7}{0.3} \approx 2.33.$$

Since $2.33 > 2.326$, z lies (just!) in the critical region: reject H_0. At the 1% level there is evidence that the fertiliser increases the mean cassava yield. Because the value is borderline, a larger sample would be advisable to confirm the result — and a larger n would reduce both α and β.

EXERCISE 6 **Practice**

1. A machine fills bags of cement labelled 50 kg; $\sigma = 0.8$ kg. A sample of 100 bags gives $\bar{x} = 49.85$ kg. Test at the 5% level whether the mean differs from 50 kg.
2. Two groups of students sat the same test. Group 1: $n_1 = 45$, $\bar{x}_1 = 13.2$, $s_1 = 2.8$; Group 2: $n_2 = 55$, $\bar{x}_2 = 12.3$, $s_2 = 3.0$. Test at the 5% level whether Group 1 performed better.
3. For question 1, state the Type I and Type II errors in context and explain which is more serious for the factory.

SOLUTION **Exercise 1**

1. Two-tailed: $H_0 : \mu = 50$, $H_1 : \mu \neq 50$; critical values ±1.96. $z = \dfrac{49.85 - 50}{0.8/10} = -1.875$. Since $|-1.875| < 1.96$, do not reject H_0: no significant evidence that the mean differs from 50 kg.

2. One-tailed: $H_0 : \mu_1 = \mu_2$, $H_1 : \mu_1 > \mu_2$; critical value 1.645. $z = \dfrac{0.9}{\sqrt{2.8^2/45 + 3.0^2/55}} = \dfrac{0.9}{\sqrt{0.1742 + 0.1636}} = \dfrac{0.9}{0.5812} \approx 1.55 < 1.645$: do not reject H_0; insufficient evidence that Group 1 performed better.
3. Type I: concluding the machine is faulty when it is not (needless, costly adjustment). Type II: missing a real fault (customers receive under- or over-filled bags). For the factory's reputation, the Type II error is arguably more serious, so a moderate α (e.g. 5%) and a reasonably large sample are sensible.

KEY POINTS **Key points of the section**

- Context dictates the test: "differs" ⇒ two-tailed; "greater/less" ⇒ one-tailed; one sample ⇒ one-mean test; two independent large samples ⇒ two-mean test.
- Choose α according to the cost of each error; a larger sample reduces both α and β.
- Follow the five steps in order — hypotheses, level and region, statistic, decision, conclusion in context — each one earns marks.
- Never conclude with symbols alone; always return to the real-world situation.

7 INTEGRATION ACTIVITY

7.1 Aims of the activity

By the end of this activity, you should be able to:

- translate a real quality-control complaint into a pair of statistical hypotheses H_0 and H_1, choosing correctly between a one-tailed and a two-tailed test;
- compute a standardised test statistic from sample data and compare it with critical values read from the standard normal table;
- carry out the complete five-step testing procedure at the 5% and 1% levels of significance and explain why the two conclusions may differ;
- interpret Type I and Type II errors in context and discuss which error is more costly for the producer and for the consumer;
- perform a two-sample z-test for the difference between two means using large independent samples;
- present and defend a statistical argument in front of the class, using precise GCE Advanced Level vocabulary.

This activity mobilises the whole chapter: types of hypotheses, significance levels, test statistics, critical regions, errors and the testing procedure, applied to both one-sample and two-sample situations.

KEY POINTS **The five-step testing procedure — reminder**

1. State the hypotheses H_0 and H_1 and decide whether the test is one-tailed or two-tailed.
2. Choose the significance level α (before looking at the data) and identify the appropriate test statistic and its distribution under H_0.
3. Determine the critical value(s) and the critical region.
4. Compute the value of the test statistic from the sample.
5. Decide and conclude: reject H_0 if the statistic falls in the critical region; otherwise do not reject H_0. State the conclusion in the language of the original problem.

7.2 Situation

The **Top Ananas Juice Company**, based in Nkongsamba (Littoral Region), bottles pineapple juice in bottles labelled "50 cl". The filling machine is set to deliver a mean of $\mu_0 = 50$ cl per bottle. From long experience, the filling process is known to produce contents that are approximately

normally distributed.

Recently, several shopkeepers in Douala have complained that the bottles seem “lighter than before”. The company’s quality-control officer, Mrs Etame, therefore selects a random sample of $n = 40$ bottles from one day’s production and measures the content of each bottle. The raw measurements (in cl) are displayed on the flipchart:

49.8	50.2	49.1	49.6	50.4	49.3	49.7	50.1
48.9	49.5	50.0	49.4	49.9	48.7	49.6	50.3
49.2	49.8	50.1	49.0	49.7	49.5	50.5	49.3
50.0	49.4	49.9	48.8	49.6	50.2	49.1	49.8
49.5	50.3	49.2	49.7	48.6	49.9	50.1	49.4

A calculator check gives the summary statistics:

$$n = 40, \qquad \bar{x} = 49.6 \text{ cl}, \qquad s = 1.2 \text{ cl}.$$

The management must decide whether the complaints are justified. If the machine is genuinely under-filling, the company risks sanctions from the consumer-protection authorities and loss of customers. If the machine is stopped and recalibrated unnecessarily, the company loses a full day of production, worth several hundred thousand francs CFA.

Three months later, a **new filling machine** is installed. A random sample of $n_2 = 50$ bottles from the new machine gives:

$$\bar{x}_2 = 50.1 \text{ cl}, \qquad s_2 = 1.4 \text{ cl}.$$

For the old machine we keep the original sample: $n_1 = 40$, $\bar{x}_1 = 49.6$ cl, $s_1 = 1.2$ cl. The company wants to know whether the two machines differ significantly in their mean fill.

Work in groups of four. Each group writes its full solution on a flipchart and presents it; the class then debates the choice of the significance level α.

7.3 Tasks

Part A — Is the old machine under-filling?

1. State, with a justification, the null hypothesis H_0 and the alternative hypothesis H_1 for Mrs Etame’s investigation. Is this test one-tailed or two-tailed? Explain your choice in the context of the shopkeepers’ complaints.

2. Explain why a z-test (standard normal) is appropriate here, even though the population standard deviation is unknown.

3. Choose the significance level $\alpha = 5\%$. Write down the critical value(s) and the critical region.

4. Compute the test statistic $z = \dfrac{\bar{x} - \mu_0}{s/\sqrt{n}}$ and state your statistical decision.

5. Write a conclusion in plain language that the managing director (who is not a statistician) can understand.

6. Repeat steps 3 to 5 at the significance level $\alpha = 1\%$. Compare the two conclusions and explain why they differ.

7. Define Type I and Type II errors in the context of this problem. For the company, what is the cost of each error? For the consumers, which error is more harmful? Discuss whether a small or a large α protects consumers better.

Part B — Do the two machines differ?

8. Formulate H_0 and H_1 to test whether the mean fills of the two machines differ. Is this test one-tailed or two-tailed?

9. Compute the two-sample test statistic

$$z = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

and carry out the test at the 5% level.

10. State your conclusion in context and suggest one practical recommendation to the management.

Part C — Presentation and debate

11. Present your five-step solutions on the flipchart. Be ready to defend: the direction of the test in Part A, the choice of α, and the interpretation of the errors.

12. Class debate: "For a juice company, is it better to use $\alpha = 1\%$ or $\alpha = 5\%$?" Each group must argue from the point of view assigned by the teacher: the company, the consumers, or the quality-control officer.

7.4 Model answer

Part A.

1. Hypotheses. The complaints concern under-filling, so we test

$$H_0 : \mu = 50 \text{ cl} \qquad \text{against} \qquad H_1 : \mu < 50 \text{ cl}.$$

This is a **one-tailed** (left-tailed) test, because only values of $\bar{x}$ below 50 provide evidence against H_0. A sample mean above 50 cl would certainly not support the shopkeepers' complaint, so it must not count as evidence against H_0.

2. Choice of test. The sample size $n = 40$ is large ($n \geq 30$), so by the Central Limit Theorem $\bar{X}$ is approximately normal and s is a reliable estimate of σ. Hence, under H_0,

$$Z = \frac{\bar{X} - \mu_0}{S/\sqrt{n}} \approx N(0,1).$$

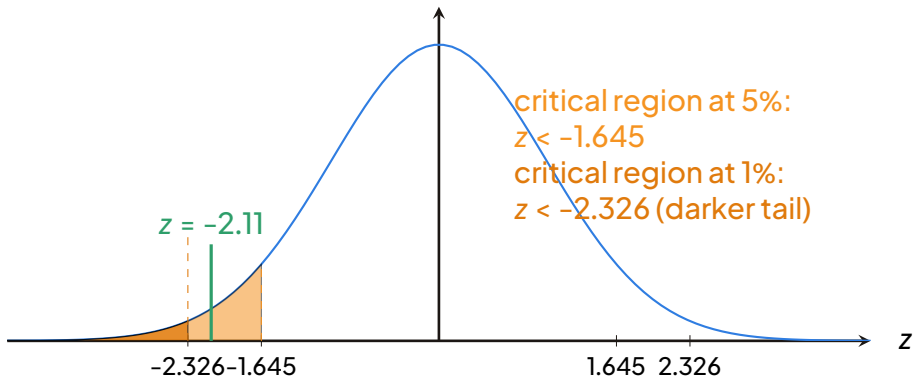
3. Critical region at $\alpha = 5\%$. For a left-tailed test at the 5% level, the whole 5% is placed in the left tail, so the critical value is -1.645 (read from the standard normal table: $\Phi(1.645) = 0.95$). The critical region is $z < -1.645$.

4. Test statistic.

$$z = \frac{\bar{x} - \mu_0}{s/\sqrt{n}} = \frac{49.6 - 50}{1.2/\sqrt{40}} = \frac{-0.4}{0.1897} \approx -2.11.$$

5. Decision and conclusion at 5%. Since $-2.11 < -1.645$, the test statistic lies in the critical region: we **reject** H_0 at the 5% level. In plain language: "There is significant evidence, at the 5% level, that the machine is filling less than 50 cl on average. The complaints appear justified."

6. Test at $\alpha = 1\%$. The critical value is now -2.326 (since $\Phi(2.326) = 0.99$). Since $-2.11 > -2.326$, the statistic is not in the critical region: we **do not reject** H_0 at the 1% level. The conclusions differ because lowering α shrinks the critical region: we demand stronger evidence before rejecting H_0. The same data can be "significant at 5%" but "not significant at 1%".



Left-tailed test for the juice machine: the observed statistic $z = -2.11$ falls inside the 5% critical region but outside the 1% critical region.

7. Errors in context.

- **Type I error** (reject H_0 when it is true): the machine is actually filling correctly, but we conclude it under-fills. Cost to the company: an unnecessary stop and recalibration, losing a day's production. Its probability is at most α.
- **Type II error** (fail to reject H_0 when it is false): the machine really under-fills, but we do nothing. Cost: consumers are systematically short-changed, and the company risks sanctions and loss of reputation.

For consumers, the Type II error is more harmful. A larger α makes it easier to reject H_0, so it reduces the risk of a Type II error: consumers are better protected by $\alpha = 5\%$ than by $\alpha = 1\%$, at the price of more false alarms for the company. Note that for a fixed sample size, decreasing α increases the probability of a Type II error; the only way to reduce both risks simultaneously is to increase the sample size.

Part B.

8. Hypotheses. We ask whether the machines differ, with no direction specified in advance:

$$H_0 : \mu_1 = \mu_2 \qquad \text{against} \qquad H_1 : \mu_1 \neq \mu_2,$$

a **two-tailed** test.

9. Test statistic. Both samples are large ($n_1 = 40 \geq 30$, $n_2 = 50 \geq 30$) and independent, so the

two-sample z-statistic is appropriate:

$$z = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}} = \frac{49.6 - 50.1}{\sqrt{\frac{1.2^2}{40} + \frac{1.4^2}{50}}} = \frac{-0.5}{\sqrt{0.036 + 0.0392}} = \frac{-0.5}{0.2742} \approx -1.82.$$

At $\alpha = 5\%$ two-tailed, the 5% is split equally between the two tails (2.5% each), so the critical values are ± 1.96. Since $-1.96 < -1.82 < 1.96$, the statistic is not in the critical region.

10. Conclusion. We **do not reject** H_0 at the 5% level: there is no significant evidence that the two machines differ in mean fill. Recommendation: the apparent improvement of the new machine is not statistically established; management should keep monitoring both machines with regular samples rather than concluding that the problem is solved.

Part C. The debate should bring out that the choice of α is a social decision, not a purely mathematical one: it balances the producer's losses against consumer protection, and it must be fixed before looking at the data.

WARNING **Common mistakes to avoid during the activity**

- Do not use a two-tailed test in Part A: the complaint specifies the direction (underfilling).
- Never write "we accept H_0"; write "we do not reject H_0 at the α% level". Failing to reject H_0 does not prove that H_0 is true.
- The significance level must be chosen before computing the statistic, not adjusted afterwards to obtain the desired conclusion.
- Do not confuse the sample standard deviation s with the standard error $s/\sqrt{n}$ in the denominator.
- In a two-tailed test, remember to split α between the two tails: at $\alpha = 5\%$ the critical values are ± 1.96, not ± 1.645.
- In the two-sample statistic, the denominator contains $\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}$ (variances divided by sample sizes), not $\frac{s_1}{\sqrt{n_1}} + \frac{s_2}{\sqrt{n_2}}$.

DID YOU KNOW? **Why "do not reject" rather than "accept"?**

A hypothesis test is like a trial: H_0 is "innocent until proven guilty". A verdict of "not guilty" does not mean the court has proved innocence — only that the evidence was insufficient to convict. In the same way, failing to reject H_0 at the 1% level in Part A does not prove the machine fills correctly; it only means the evidence was not strong enough at that level. This is why the same sample can acquit at 1% and convict at 5%.

8 METHODS AND WORKED EXAMPLES

8.1 Method 1: Setting Up Hypotheses and Choosing the Type of Test

A **statistical hypothesis** is a claim about a population parameter (for example the mean μ). Because we can only observe a sample, we can never prove such a claim directly; instead, we weigh the sample evidence against a default position. This is the logic of every **hypothesis test** (also called a **significance test**).

DEFINITION **Null and alternative hypotheses**

The **null hypothesis** H_0 is the statement assumed to be true unless the data provide strong evidence against it; it always contains an equality. The **alternative hypothesis** H_1 is the statement accepted when H_0 is rejected.

METHOD **Formulating H_0 and H_1**

To translate a practical question into a statistical test:

1. Identify the **parameter** involved (mean μ, proportion p, difference of means $\mu_1 - \mu_2$).
2. Write the **null hypothesis** H_0: it always contains an equality (e.g. $\mu = \mu_0$). It represents the status quo, the claim assumed true until the data prove otherwise.
3. Write the **alternative hypothesis** H_1 from the wording of the question:
   - "different from", "changed", "not equal to" $\Rightarrow H_1 : \mu \neq \mu_0$ (**two-tailed** test);
   - "greater than", "increased", "more than" $\Rightarrow H_1 : \mu > \mu_0$ (**one-tailed, right**);
   - "less than", "decreased", "fewer than" $\Rightarrow H_1 : \mu < \mu_0$ (**one-tailed, left**).
4. State the **level of significance** α (given in the question, often 5% or 1%).

Reflex: the claim the investigator wants to establish usually goes into H_1, because a test can only give strong evidence against H_0.

WORKED EXAMPLE 12 **Choosing hypotheses for a maize yield study**

An agricultural station near Bafoussam has grown a variety of maize whose mean yield is known to be 2.4 tonnes per hectare. A new fertiliser is tried on 50 plots. The agronomist wishes to test whether the fertiliser increases the mean yield. Write down suitable null and alternative hypotheses, state the type of test, and explain what a test at the 5% level of significance means.

SOLUTION

Solution. The parameter is the population mean yield μ (tonnes per hectare) with the new fertiliser.
The status quo is "the fertiliser makes no difference", so

$$H_0 : \mu = 2.4.$$

The agronomist wants to establish an increase, so

$$H_1 : \mu > 2.4.$$

This is a one-tailed (right-tailed) test at level $\alpha = 0.05$.
Interpretation of the 5% level: if H_0 is true, the probability of obtaining a sample result extreme enough to reject H_0 is at most 0.05. In other words, we accept a 5% risk of wrongly concluding that the fertiliser works when in fact it does not (a Type I error).

8.2 Method 2: Finding Critical Values and the Critical Region

The **test statistic** measures how far the sample result lies from the value claimed by H_0, in units of the standard error. For tests on means with σ known or n large, the test statistic follows the standard normal distribution $N(0,1)$ under H_0 (by the Central Limit Theorem). The **critical value**(s) delimit the **critical region**: the set of values of the test statistic so extreme that, if H_0 were true, they would occur with probability at most α.

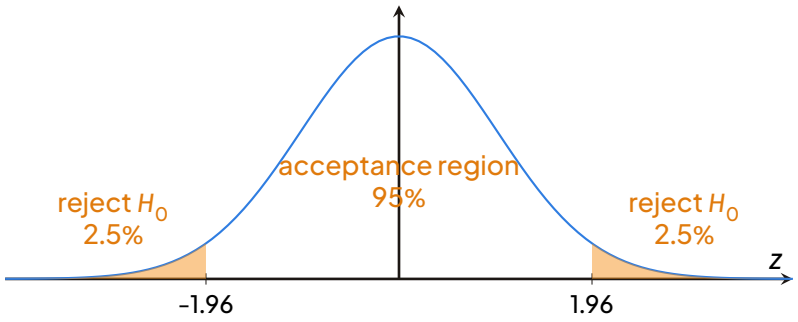
METHOD **Critical values for a z-test**

For a test based on the standard normal distribution $Z \sim N(0,1)$:

1. Read the significance level α and the type of test.
2. Determine the **critical value**(s):

Type of test	Rejection condition	Critical value(s)
Two-tailed ($H_1 : \mu \neq \mu_0$)	$\lvert z \rvert > z_{\alpha/2}$	±1.960 at 5%; ±2.576 at 1%
Right-tailed ($H_1 : \mu > \mu_0$)	$z > z_{\alpha}$	1.645 at 5%; 2.326 at 1%
Left-tailed ($H_1 : \mu < \mu_0$)	$z < -z_{\alpha}$	-1.645 at 5%; -2.326 at 1%

3. The **critical region** (rejection region) is the set of values of the test statistic beyond the critical value(s); the **acceptance region** is its complement.
4. Reflex: for a two-tailed test, split α equally between the two tails ($\alpha/2$ each).



Two-tailed test at the 5% level: the critical region consists of the two tails beyond ±1.96.

WORKED EXAMPLE 13 **Critical region in terms of the sample mean**

A machine fills sugar packets with nominal mass $\mu_0 = 500$ g. The mass is known to be normally distributed with standard deviation $\sigma = 8$ g. A sample of $n = 64$ packets is taken. A two-tailed test of $H_0 : \mu = 500$ against $H_1 : \mu \neq 500$ is carried out at the 5% level. Find the critical region in terms of the sample mean $\bar{X}$.

SOLUTION

Solution. Under H_0, $\bar{X} \sim N\left(500, \dfrac{8^2}{64}\right)$, i.e. standard error

$$\sigma_{\bar{X}} = \frac{\sigma}{\sqrt{n}} = \frac{8}{\sqrt{64}} = \frac{8}{8} = 1\,\text{g}.$$

Two-tailed test at 5%: critical values ±1.960. We reject H_0 when

$$\left|\frac{\bar{X} - 500}{1}\right| > 1.960 \quad \Longleftrightarrow \quad \bar{X} < 500 - 1.96 \text{ or } \bar{X} > 500 + 1.96.$$

Hence the critical region is

$$\bar{X} < 498.04\,\text{g} \quad \text{or} \quad \bar{X} > 501.96\,\text{g},$$

and the acceptance region is $498.04 \le \bar{X} \le 501.96$ g.
Check: converting a sample mean to a decision is often easier in "original units"; always convert the z-critical values back using $\bar{x} = \mu_0 \pm z_{\alpha/2}\,\sigma/\sqrt{n}$.

8.3 Method 3: One-Sample z-Test for a Population Mean (σ Known or n Large)

METHOD **The five-step testing procedure**

1. State H_0 and H_1 and the level α.
2. Compute the test statistic
$$z = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}} \qquad \text{(use } s \text{ for } \sigma \text{ if } \sigma \text{ unknown and } n \text{ large).}$$
3. Determine the critical value(s) from $N(0,1)$ tables.
4. Decide: reject H_0 if z falls in the critical region; otherwise do not reject H_0.
5. Conclude in the language of the problem, mentioning the level of significance.

Reflex: "do not reject H_0" does not prove H_0 true; it means the evidence is insufficient against it.

WORKED EXAMPLE 14 **Testing the mean lifetime of bulbs**

A manufacturer in Douala claims that the mean lifetime of its electric bulbs is 1200 hours. Past experience gives a standard deviation of $\sigma = 120$ hours. A random sample of 36 bulbs has a mean lifetime of 1160 hours. Test, at the 5% level of significance, whether the mean lifetime is less than 1200 hours.

SOLUTION

Solution.
Step 1 — Hypotheses. Left-tailed test:
$$H_0 : \mu = 1200 \qquad H_1 : \mu < 1200, \qquad \alpha = 0.05.$$
Step 2 — Test statistic.
$$z = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}} = \frac{1160 - 1200}{120/\sqrt{36}} = \frac{-40}{20} = -2.00.$$
Step 3 — Critical value. Left-tailed at 5%: critical value -1.645; critical region $z < -1.645$.
Step 4 — Decision. Since $-2.00 < -1.645$, the test statistic lies in the critical region: we reject H_0.
Step 5 — Conclusion. At the 5% level of significance, there is sufficient evidence that the mean lifetime of the bulbs is less than 1200 hours; the manufacturer's claim is not supported by the data.
Remark (p-value): $P(Z < -2.00) = 0.0228 < 0.05$, confirming the rejection. The **p-value** is the probability, under H_0, of a result at least as extreme as the one observed; we reject H_0 exactly when the p-value is less than α.

8.4 Method 4: Two-Tailed Test with Unknown Variance (Large Sample)

METHOD **Using the sample standard deviation**

When σ is unknown but $n \geq 30$:

1. Estimate σ by the sample standard deviation s (use the unbiased estimate $\hat{\sigma}^2 = \dfrac{n}{n-1}s_n^2$ if only the biased variance is given).
2. Compute $z = \dfrac{\bar{x} - \mu_0}{s/\sqrt{n}}$ and compare with the two-tailed critical values $\pm z_{\alpha/2}$.
3. Conclude in context.

Trap to avoid: do not confuse the sample standard deviation with the standard error $s/\sqrt{n}$ — forgetting $\sqrt{n}$ is the most common error in the GCE.

WORKED EXAMPLE 15 **Mean score in a mock examination**

The national mean mark in a mathematics mock examination has been 52 over the years. A sample of 100 candidates from the Centre Region gives a mean of 54.2 with standard deviation 12.5. Test at the 1% level whether the mean mark of candidates from this region differs from the national mean.

SOLUTION

Solution.
Step 1. Two-tailed test:

$$H_0 : \mu = 52, \qquad H_1 : \mu \neq 52, \qquad \alpha = 0.01.$$

Step 2. σ unknown, $n = 100$ large, so use $s = 12.5$:

$$z = \frac{\bar{x} - \mu_0}{s/\sqrt{n}} = \frac{54.2 - 52}{12.5/\sqrt{100}} = \frac{2.2}{1.25} = 1.76.$$

Step 3. Critical values at 1% (two-tailed): ± 2.576.
Step 4. Since $|1.76| = 1.76 < 2.576$, we do not reject H_0.
Step 5. At the 1% level, there is insufficient evidence that the mean mark in the Centre Region differs from the national mean of 52.
Remark: at the 5% level (critical values ± 1.96) we would still not reject, since $1.76 < 1.96$. The conclusion is robust to the choice between these two levels.

8.5 Method 5: Type I and Type II Errors

Whatever decision we take, it may be wrong, because it is based on a sample. There are exactly two ways of being wrong, and it is essential to distinguish them.

DEFINITION **The two types of error**

A **Type I error** consists in rejecting H_0 when H_0 is in fact true. A **Type II error** consists in not rejecting H_0 when H_0 is in fact false.

	H_0 true	H_0 false
Reject H_0	Type I error (probability α)	correct decision
Do not reject H_0	correct decision	Type II error (probability β)

METHOD **Computing error probabilities**

1. **Type I error**: rejecting H_0 when it is true. Its probability is
$$P(\text{Type I}) = \alpha = P(\bar{X} \in \text{critical region} \mid H_0 \text{ true}).$$
2. **Type II error**: not rejecting H_0 when a specific alternative $\mu = \mu_1$ is true. Compute
$$\beta = P(\bar{X} \in \text{acceptance region} \mid \mu = \mu_1)$$
using the normal distribution centred at μ_1 with the same standard error.
3. The **power** of the test against μ_1 is $1 - \beta$.

Reflex: first write the critical region in terms of $\bar{X}$; then restandardise with the alternative mean for β.

WORKED EXAMPLE 16 **Type I and Type II errors for the sugar packets**

Recall the sugar-packet test: $H_0 : \mu = 500$ against $H_1 : \mu \neq 500$, $\sigma = 8$, $n = 64$, critical region $\bar{X} < 498.04$ or $\bar{X} > 501.96$.

(a) State the probability of a Type I error.

(b) If the true mean is in fact $\mu = 503$ g, calculate the probability of a Type II error and the power of the test.

SOLUTION

Solution. The standard error is $\sigma/\sqrt{n} = 1$ g.
(a) By construction, $P(\text{Type I}) = \alpha = 0.05$.
(b) Under $\mu = 503$, $\bar{X} \sim N(503, 1)$. A Type II error occurs when $\bar{X}$ falls in the acceptance

region $[498.04,\ 501.96]$:

$$\begin{aligned}\beta &= P(498.04 < \bar{X} < 501.96 \mid \mu = 503)\\ &= P\left(\frac{498.04 - 503}{1} < Z < \frac{501.96 - 503}{1}\right)\\ &= P(-4.96 < Z < -1.04)\\ &= \Phi(-1.04) - \Phi(-4.96) \approx 0.1492 - 0 = 0.1492.\end{aligned}$$

So the probability of a Type II error is about 0.149, and the power of the test against $\mu = 503$ is

$$1 - \beta \approx 0.851.$$

Interpretation: if the machine really overfills by 3 g on average, this test detects it about 85% of the time. Increasing n or α would increase the power.

WARNING **The trade-off between α and β**

For a fixed sample size, making α smaller (a stricter test) enlarges the acceptance region and therefore increases β: one cannot reduce both errors simultaneously. The only way to reduce both is to increase the sample size n, which shrinks the standard error $\sigma/\sqrt{n}$.

8.6 Method 6: Test for the Difference Between Two Means (Large Independent Samples)

To compare two populations we take a large sample from each. If the samples are independent, the difference of sample means satisfies, approximately,

$$\bar{X}_1 - \bar{X}_2 \sim N\left(\mu_1 - \mu_2,\ \frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}\right),$$

because the variance of a difference of independent variables is the sum of the variances.

METHOD **Two-sample z-test**

For two large independent samples ($n_1, n_2 \geq 30$):

1. Hypotheses: $H_0 : \mu_1 - \mu_2 = 0$ (i.e. $\mu_1 = \mu_2$) against the appropriate H_1.
2. Test statistic:
$$z = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\dfrac{s_1^2}{n_1} + \dfrac{s_2^2}{n_2}}}.$$
3. Compare $|z|$ (or z) with the critical value at level α; decide and conclude in context.

Reflexes: the standard error of a difference combines both variances; variances add, never standard deviations. Check that the samples are independent.

WORKED EXAMPLE 17 **Comparing two brands of cement**

A building contractor in Yaoundé compares the crushing strengths (in $\mathrm{N\,mm^{-2}}$) of two brands of cement. Random samples give:

Brand	n	Mean	Standard deviation
A	60	32.5	3.2
B	50	31.4	2.8

Test at the 5% level whether brand A has a greater mean crushing strength than brand B.

SOLUTION

Solution.
Step 1. Right-tailed test:

$$H_0 : \mu_A - \mu_B = 0, \qquad H_1 : \mu_A - \mu_B > 0, \qquad \alpha = 0.05.$$

Step 2. Standard error of the difference:

$$\mathrm{SE} = \sqrt{\frac{3.2^2}{60} + \frac{2.8^2}{50}} = \sqrt{\frac{10.24}{60} + \frac{7.84}{50}} = \sqrt{0.1707 + 0.1568} = \sqrt{0.3275} \approx 0.5723.$$

Test statistic:

$$z = \frac{32.5 - 31.4}{0.5723} = \frac{1.1}{0.5723} \approx 1.92.$$

Step 3. Critical value (right-tailed, 5%): 1.645.
Step 4. Since $1.92 > 1.645$, we reject H_0.
Step 5. At the 5% level of significance, the samples provide sufficient evidence that the mean crushing strength of brand A is greater than that of brand B.
Remark (p-value): $P(Z > 1.92) \approx 0.027 < 0.05$, consistent with rejection; note that at the 1% level we would not reject, since $1.92 < 2.326$.

8.7 Method 7: Two-Tailed Test for a Difference and Confidence-Interval Link

METHOD **Full two-sample procedure with interpretation**

1. State $H_0 : \mu_1 = \mu_2$ against $H_1 : \mu_1 \neq \mu_2$; choose α.

2. Compute the standard error $\sqrt{s_1^2/n_1 + s_2^2/n_2}$ and z.
3. Decide using $\pm z_{\alpha/2}$.
4. Link: a two-tailed test at level α rejects $H_0 : \mu_1 = \mu_2$ exactly when the $(1-\alpha)$ confidence interval for $\mu_1 - \mu_2$,

$$(\bar{x}_1 - \bar{x}_2) \pm z_{\alpha/2}\,\mathrm{SE},$$

does not contain 0.

WORKED EXAMPLE 18 **Heights of seedlings from two nurseries**

An agroforestry project measures the heights (cm) of 6-month-old seedlings from two nurseries:

Nursery	n	Mean height	Standard deviation
1	80	45.6	6.4
2	100	44.1	7.0

(a) Test at the 5% level whether the two nurseries differ in mean seedling height.

(b) Construct a 95% confidence interval for $\mu_1 - \mu_2$ and verify the link with part (a).

SOLUTION

Solution.

(a) $H_0 : \mu_1 - \mu_2 = 0$; $H_1 : \mu_1 - \mu_2 \neq 0$; $\alpha = 0.05$ (two-tailed).

$$\mathrm{SE} = \sqrt{\frac{6.4^2}{80} + \frac{7.0^2}{100}} = \sqrt{\frac{40.96}{80} + \frac{49}{100}} = \sqrt{0.512 + 0.490} = \sqrt{1.002} \approx 1.001.$$

$$z = \frac{45.6 - 44.1}{1.001} = \frac{1.5}{1.001} \approx 1.50.$$

Critical values: ± 1.960. Since $|1.50| < 1.960$, we do not reject H_0: at the 5% level there is insufficient evidence of a difference in mean height between the two nurseries.

(b) The 95% confidence interval is

$$1.5 \pm 1.96 \times 1.001 = 1.5 \pm 1.96 = (-0.46,\ 3.46)\ \mathrm{cm}.$$

Since this interval contains 0, the value $\mu_1 - \mu_2 = 0$ is plausible — exactly the conclusion of the test in (a). The two approaches agree, as they always must for a two-tailed z-test at the matching level.

KEY POINTS **Key points of the chapter**

- H_0 always contains an equality; the claim to be established goes into H_1.
- Two-tailed: split α into $\alpha/2$ in each tail (critical values ±1.960 at 5%, ±2.576 at 1%). One-tailed: ±1.645 at 5%, ±2.326 at 1%.
- Test statistic for one mean: $z = \dfrac{\bar{x} - \mu_0}{\sigma/\sqrt{n}}$; for a difference of two means (large independent samples): $z = \dfrac{\bar{x}_1 - \bar{x}_2}{\sqrt{s_1^2/n_1 + s_2^2/n_2}}$.
- $P(\text{Type I error}) = \alpha$; β is computed with the distribution under a specified alternative; power $= 1 - \beta$.
- Always finish with a conclusion in context, quoting the level of significance.

WARNING **Frequent errors in the GCE examination**

- Writing H_1 with an equality sign, or putting the claim to be proved in H_0.
- Using α instead of $\alpha/2$ for a two-tailed test (critical value 1.645 instead of 1.960).
- Forgetting $\sqrt{n}$ in the standard error, or adding standard deviations instead of variances in two-sample tests.
- Concluding "H_0 is true" instead of "there is insufficient evidence to reject H_0".
- Giving a decision without a conclusion in the context of the problem (units and situation must appear).

9 EXERCISES

9.1 I check my knowledge

EXERCISE 7 **MCQ: vocabulary of testing**

For each question, choose the single correct answer, justifying briefly.

1. In a test of $H_0 : \mu = 50$ against $H_1 : \mu \neq 50$ at the 5% level (normal model, σ known), the critical values of the standardised statistic are:

   (a) ± 1.645 (b) ± 1.960 (c) ± 2.576 (d) ± 2.326

2. The significance level α of a test is:

   (a) the probability of accepting H_0 when H_0 is true;

   (b) the probability of rejecting H_0 when H_0 is true;

   (c) the probability of accepting H_0 when H_0 is false;

   (d) the probability of rejecting H_0 when H_0 is false.

3. For a one-tailed test of $H_0 : \mu = \mu_0$ against $H_1 : \mu > \mu_0$ at the 1% level, the critical value of z is:

   (a) 1.645 (b) 1.960 (c) 2.326 (d) 2.576

4. If the computed value of the test statistic falls in the critical region, we:

   (a) accept H_0; (b) reject H_0 in favour of H_1; (c) increase α; (d) take a larger sample.

EXERCISE 8 **True or false?**

Say whether each statement is true or false, justifying your answer carefully.

1. Decreasing the significance level α makes the critical region larger.
2. A Type I error consists in rejecting H_0 when H_0 is actually true.
3. The probability of a Type I error is always equal to the significance level α.
4. In a two-tailed test at the 5% level, each tail of the critical region carries probability 0.05.
5. For a large sample, the sample standard deviation s may be used in place of an unknown population standard deviation σ.

EXERCISE 9 **Definitions**

Define each of the following terms precisely, in your own words, and illustrate each with a short example:

1. null hypothesis and alternative hypothesis;
2. one-tailed test and two-tailed test;
3. test statistic;
4. critical region and critical value;
5. Type I error and Type II error.

EXERCISE 10 **Direct calculations**

1. A sample of $n = 36$ observations from a population with known standard deviation $\sigma = 12$ gives $\bar{x} = 104$. Compute the value of the standardised test statistic z for testing $H_0 : \mu = 100$.
2. For a two-tailed test of $H_0 : \mu = 100$ at the 5% level with $n = 36$ and $\sigma = 12$, find the critical values of $\bar{X}$ and state the acceptance region.
3. State the critical values of z for: (i) a one-tailed test at 5%; (ii) a two-tailed test at 1%.

9.2 I practise

EXERCISE 11 **The miller's flour bags**

A miller in Bamenda claims that his bags of flour weigh 25 kg on average. A random sample of 50 bags gives a mean weight $\bar{x} = 24.8$ kg with standard deviation $s = 1.5$ kg. Test, at the 5% level of significance, whether the mean weight is less than 25 kg. State your hypotheses, the test statistic, the critical value, and your conclusion in context.

EXERCISE 12 **GCE performance**

The mean score of a school in last year's GCE was 11.2. This year, a random sample of 60 candidates has mean score 11.9 with standard deviation 3.1. Test, at the 1% level, whether the performance of the school has improved. Explain why a one-tailed test is appropriate here.

EXERCISE 13 **Two varieties of cassava**

Two varieties of cassava are compared at an experimental farm near Ebolowa. A sample of 40 cuttings of variety A gives a mean yield of 18.2 t/ha with standard deviation 2.4 t/ha, while 45 cuttings of variety B give a mean yield of 17.1 t/ha with standard deviation 2.8 t/ha. Test, at the 5% level, whether the mean yields of the two varieties differ.

EXERCISE 14 **A new drug**

A hospital finds that a sample of 35 patients treated with a new drug recover in a mean time of 6.2 days (standard deviation 1.8 days), compared with a known mean recovery time of 7.0 days for the standard drug.

1. Test, at the 1% level, whether the new drug leads to a faster recovery.
2. Comment on the choice of a one-tailed rather than a two-tailed test in this situation.

9.3 I prepare for the GCE

EXERCISE 15 **Errors and confidence intervals**

1. Explain, with the help of a concrete example, the difference between a Type I error and a Type II error in hypothesis testing, and state the probability of a Type I error when the significance level is $\alpha = 0.05$. (6 marks)
2. A 95% confidence interval for the mean content of bottles filled by a machine is $(48.9, 50.3)$ cl. Without further calculation, decide the outcome of the two-tailed test of $H_0 : \mu = 50$ against $H_1 : \mu \neq 50$ at the 5% level, justifying your answer. (4 marks)
3. Would the same conclusion hold for the test of $H_0 : \mu = 50.5$? Justify. (3 marks)

EXERCISE 16 **Comparing two schools**

Two independent random samples of GCE candidates from schools X and Y give the following results: school X: $n_1 = 50$, $\bar{x}_1 = 12.4$, $s_1 = 3.0$; school Y: $n_2 = 55$, $\bar{x}_2 = 11.5$, $s_2 = 3.4$.

1. Test, at the 5% level, the hypothesis that the two schools have the same mean score against the hypothesis that the means differ. (7 marks)
2. Test, at the 5% level, the claim that school X's mean exceeds school Y's mean by more than 0.5. State clearly the null and alternative hypotheses. (6 marks)

3. State one assumption you have made about the samples, and explain why the use of the normal distribution is justified here. (3 marks)

EXERCISE 17 **Essay: designing and reporting a test**

A cocoa cooperative in the Centre Region suspects that a new fermentation method changes the mean weight of dried cocoa beans per pod, previously known to be $\mu_0 = 38.0$ g. A random sample of 64 pods processed by the new method gives $\bar{x} = 39.1$ g and $s = 4.8$ g.

1. Write a short essay describing the complete procedure of a hypothesis test: choice of hypotheses, choice between a one-tailed and a two-tailed test (justify your choice), significance level, test statistic, critical region, decision rule, and conclusion. (8 marks)

2. Carry out the test you have described at the 5% level and state your conclusion in the context of the cooperative. (6 marks)

3. Explain what a Type II error would mean for the cooperative in this context, and state one way of reducing its probability. (4 marks)

10 SOLUTIONS TO THE EXERCISES

SOLUTION **Exercise 1 — MCQ: vocabulary of testing**

1. Answer: (b) ±1.960. The test is two-tailed ($H_1 : \mu \neq 50$) at the 5% level, so each tail carries probability $\alpha/2 = 0.025$. We need z with $P(Z > z) = 0.025$, i.e. $\Phi(z) = 0.975$, giving $z = 1.960$. The critical values are ±1.960.
2. Answer: (b). By definition, the significance level is the probability of rejecting H_0 when H_0 is true, i.e. $\alpha = P(\text{reject } H_0 \mid H_0 \text{ true}) = P(\text{Type I error})$.
3. Answer: (c) 2.326. For a one-tailed (right-tailed) test at the 1% level, the whole probability 0.01 lies in the right tail: $\Phi(z) = 0.99$ gives $z = 2.326$.
4. Answer: (b). The critical (rejection) region is precisely the set of values of the test statistic that lead to the rejection of H_0 in favour of H_1.

SOLUTION **Exercise 2 — True or false?**

1. False. Decreasing α (e.g. from 5% to 1%) pushes the critical values further into the tails (1.960 → 2.576 two-tailed), so the critical region becomes smaller: we demand stronger evidence before rejecting H_0.
2. True. This is the definition: a Type I error is rejecting H_0 when H_0 is actually true (a "false alarm").
3. True. By construction of the test, $P(\text{Type I error}) = P(\text{reject } H_0 \mid H_0 \text{ true}) = \alpha$. (More precisely, for composite H_0 it is the supremum over H_0, which equals α at the boundary.)
4. False. In a two-tailed test the total probability $\alpha = 0.05$ is shared equally between the two tails, so each tail carries probability 0.025, not 0.05.
5. True. For large n (in practice $n \geq 30$), s is a reliable estimate of σ, and by the Central Limit Theorem $Z = \dfrac{\bar{X} - \mu_0}{s/\sqrt{n}}$ is approximately $N(0,1)$ under H_0.

SOLUTION **Exercise 3 — Definitions**

1. Null hypothesis H_0: the statement about a population parameter that is assumed true

and is placed on trial; it always contains an equality. Alternative hypothesis H_1: the statement accepted if H_0 is rejected. Example: $H_0 : \mu = 25$ kg against $H_1 : \mu < 25$ kg for the miller's bags.

2. One-tailed test: H_1 specifies a direction ($\mu > \mu_0$ or $\mu < \mu_0$); the critical region lies in a single tail. Two-tailed test: H_1 states only a difference ($\mu \neq \mu_0$); the critical region is split between both tails. Example: testing whether a mean "has changed" is two-tailed; testing whether it "has increased" is one-tailed.

3. Test statistic: a function of the sample data, with known distribution under H_0, used to decide. Example: $Z = \dfrac{\bar{X} - \mu_0}{\sigma/\sqrt{n}} \sim N(0,1)$ under H_0.

4. Critical region: the set of values of the test statistic leading to rejection of H_0; critical value(s): the boundary value(s) of that region. Example: critical region $|z| > 1.960$ at the 5% two-tailed level; critical values ± 1.960.

5. Type I error: rejecting H_0 when it is true; its probability is α. Type II error: accepting H_0 when it is false; its probability is denoted β. Example: convicting an innocent person (Type I) versus acquitting a guilty one (Type II).

SOLUTION **Exercise 4 — Direct calculations**

1. Under $H_0 : \mu = 100$, with $\sigma = 12$ and $n = 36$:

$$z = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}} = \frac{104 - 100}{12/\sqrt{36}} = \frac{4}{2} = 2.0.$$

2. The critical values of z at the 5% two-tailed level are ± 1.960. Converting to $\bar{X}$:

$$\bar{x} = 100 \pm 1.960 \times \frac{12}{\sqrt{36}} = 100 \pm 3.92,$$

so the critical values of $\bar{X}$ are 96.08 and 103.92. The acceptance region (non-rejection region) is $96.08 < \bar{x} < 103.92$; the critical region is $\bar{x} < 96.08$ or $\bar{x} > 103.92$.

3. (i) One-tailed at 5%: $z = 1.645$ (or -1.645 for a left tail). (ii) Two-tailed at 1%: each tail carries 0.005, $\Phi(z) = 0.995$, so $z = \pm 2.576$.

SOLUTION **Exercise 5 — The miller's flour bags**

Hypotheses. We suspect under-filling, so a left-tailed test:

$$H_0 : \mu = 25\text{ kg} \qquad \text{against} \qquad H_1 : \mu < 25\text{ kg}.$$

Test statistic. Since $n = 50$ is large, s estimates σ and under H_0:

$$Z = \frac{\bar{X} - 25}{s/\sqrt{n}} \approx N(0,1), \qquad z = \frac{24.8 - 25}{1.5/\sqrt{50}} = \frac{-0.2}{0.2121} \approx -0.94.$$

Critical value. One-tailed at 5%: $z_{\text{crit}} = -1.645$; critical region $z < -1.645$. Decision. Since $-0.94 > -1.645$, the statistic does not fall in the critical region: we do not reject H_0. Conclusion. At the 5% level, the sample does not provide sufficient evidence that the mean weight of the miller's bags is less than 25 kg.

SOLUTION **Exercise 6 — GCE performance**

Why one-tailed? The question asks whether performance has improved, i.e. only an increase is of interest; a decrease would not support the claim. Hence a right-tailed test:

$$H_0 : \mu = 11.2 \qquad \text{against} \qquad H_1 : \mu > 11.2.$$

Test statistic. With $n = 60$ (large), $s = 3.1$ estimates σ:

$$z = \frac{11.9 - 11.2}{3.1/\sqrt{60}} = \frac{0.7}{0.4002} \approx 1.75.$$

Critical value. One-tailed at 1%: $z_{\text{crit}} = 2.326$. Decision. Since $1.75 < 2.326$, we do not reject H_0. Conclusion. At the 1% level there is insufficient evidence that the school's mean GCE score has improved. (Note: the improvement would have been declared significant at the 5% level, since $1.75 > 1.645$ — the 1% level demands stronger evidence.)

SOLUTION **Exercise 7 — Two varieties of cassava**

Hypotheses. We test for any difference (two-tailed):

$$H_0 : \mu_A = \mu_B \qquad \text{against} \qquad H_1 : \mu_A \neq \mu_B.$$

Test statistic. Both samples are large, so under H_0:

$$Z = \frac{\bar{X}_A - \bar{X}_B}{\sqrt{\dfrac{s_A^2}{n_A} + \dfrac{s_B^2}{n_B}}} \approx N(0,1).$$

Compute the standard error:

$$\sqrt{\frac{2.4^2}{40}+\frac{2.8^2}{45}}=\sqrt{\frac{5.76}{40}+\frac{7.84}{45}}=\sqrt{0.144+0.1742}=\sqrt{0.3182}\approx 0.564.$$

Hence

$$z=\frac{18.2-17.1}{0.564}=\frac{1.1}{0.564}\approx 1.95.$$

Critical values. Two-tailed at 5%: ±1.960. Decision. Since $1.95 < 1.960$ (just barely), the statistic does not fall in the critical region: we do not reject H_0. Conclusion. At the 5% level, the samples do not provide sufficient evidence of a difference between the mean yields of the two cassava varieties. The result is borderline: a slightly larger sample could change the conclusion.

SOLUTION **Exercise 8 — A new drug**

1. Hypotheses. "Faster recovery" means a smaller mean time, so a left-tailed test:

   $$H_0:\mu=7.0 \qquad \text{against} \qquad H_1:\mu<7.0.$$

   Test statistic. With $n=35$ (large), $s=1.8$ estimates σ:

   $$z=\frac{6.2-7.0}{1.8/\sqrt{35}}=\frac{-0.8}{0.3043}\approx -2.63.$$

   Critical value. One-tailed at 1%: $z_{\text{crit}}=-2.326$; critical region $z<-2.326$. Decision. Since $-2.63<-2.326$, we reject H_0. Conclusion. At the 1% level there is significant evidence that the new drug reduces the mean recovery time.

2. A one-tailed test is appropriate because the research question is directional: the hospital wants to know whether the new drug is better (faster). A two-tailed test would also count evidence that the drug is slower, diluting the rejection probability across two tails and making it harder to detect the improvement of interest. The one-tailed test concentrates all of α in the relevant tail, giving more power to detect a genuine improvement — but it must be chosen before seeing the data.

SOLUTION **Exercise 9 — Errors and confidence intervals**

1. (6 marks) A Type I error is rejecting H_0 when it is true; a Type II error is accepting H_0 when it is false. Example: a machine fills bottles with a claimed mean of 50 cl. A Type I error occurs if a test concludes the machine is faulty when it is actually correctly set (needless, costly adjustment); a Type II error occurs if the test fails to detect a real

fault (under-filled bottles reach customers). The probability of a Type I error equals the significance level: $P(\text{Type I}) = \alpha = 0.05$. Mark scheme: definition of Type I (1), definition of Type II (1), coherent example (2), statement $P(\text{Type I}) = \alpha$ (1), value 0.05 (1).

2. (4 marks) The 95% confidence interval $(48.9,\ 50.3)$ contains the value $\mu_0 = 50$. A 95% confidence interval consists exactly of the values of μ_0 that would not be rejected by a two-tailed test at the 5% level. Therefore we do not reject $H_0 : \mu = 50$ at the 5% level. Mark scheme: observation $50 \in (48.9, 50.3)$ (1), link between CI and two-tailed test (2), correct decision (1).

3. (3 marks) No. The value 50.5 lies outside the interval $(48.9,\ 50.3)$, so the two-tailed test of $H_0 : \mu = 50.5$ at the 5% level would reject H_0. Mark scheme: observation $50.5 \notin$ CI (1), correct decision (1), justification via CI–test duality (1).

SOLUTION **Exercise 10 — Comparing two schools**

1. (7 marks) Hypotheses (two-tailed):

$$H_0 : \mu_1 = \mu_2 \qquad \text{against} \qquad H_1 : \mu_1 \neq \mu_2.$$

Standard error:

$$\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}} = \sqrt{\frac{9}{50} + \frac{11.56}{55}} = \sqrt{0.18 + 0.2102} = \sqrt{0.3902} \approx 0.625.$$

Test statistic:

$$z = \frac{12.4 - 11.5}{0.625} = \frac{0.9}{0.625} = 1.44.$$

Critical values at 5% (two-tailed): ± 1.960. Since $1.44 < 1.960$, we do not reject H_0: at the 5% level there is no significant evidence that the two schools' mean scores differ. Mark scheme: hypotheses (1), formula for SE (1), SE value (1), z value (1), critical values (1), comparison and decision (1), conclusion in context (1).

2. (6 marks) Let $\delta = 0.5$. The claim is $\mu_1 - \mu_2 > 0.5$, so:

$$H_0 : \mu_1 - \mu_2 = 0.5 \qquad \text{against} \qquad H_1 : \mu_1 - \mu_2 > 0.5 \quad \text{(one-tailed)}.$$

Under H_0:

$$z = \frac{(\bar{x}_1 - \bar{x}_2) - 0.5}{0.625} = \frac{0.9 - 0.5}{0.625} = 0.64.$$

Critical value at 5% (one-tailed): 1.645. Since $0.64 < 1.645$, we do not reject H_0: the data do not support the claim that school X's mean exceeds school Y's by more than 0.5. Mark scheme: correct H_0 with equality at 0.5 (2), one-tailed H_1 (1), z calculation (1), critical value (1), decision and conclusion (1).

3. (3 marks) Assumption: the two samples are independent and randomly drawn. Use of the normal distribution is justified because both samples are large ($n_1 = 50$, $n_2 = 55 \geq 30$): by the Central Limit Theorem $\bar{X}_1 - \bar{X}_2$ is approximately normal, and s_1, s_2 are reliable estimates of σ_1, σ_2. Mark scheme: independence (1), CLT for large samples (1), $s \approx \sigma$ (1).

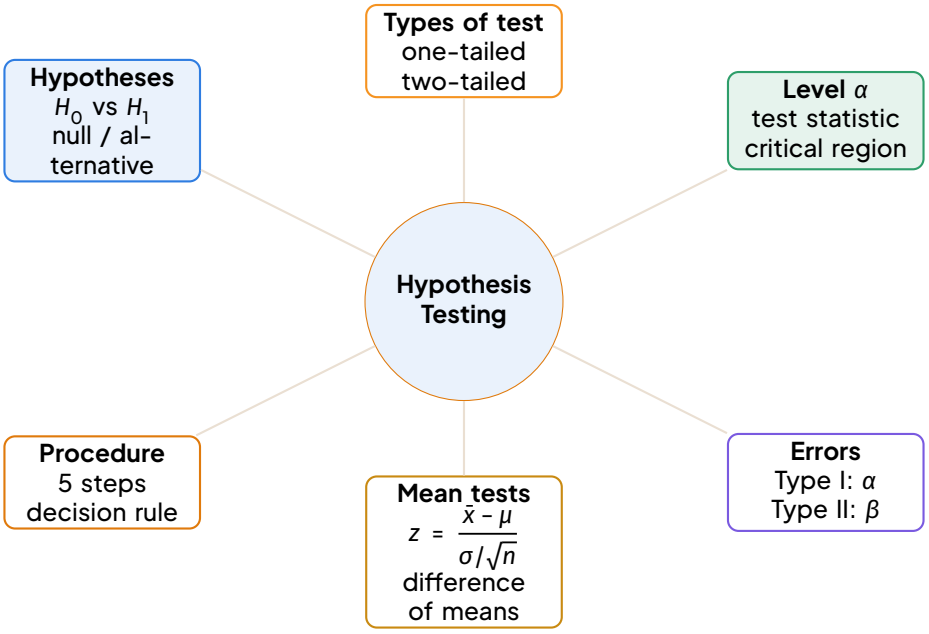
SOLUTION **Exercise 11 — Essay: designing and reporting a test**

1. (8 marks) Procedure. Step 1 — Hypotheses: state $H_0 : \mu = \mu_0$ (the status quo, with equality) and H_1. Step 2 — Choice of test: the cooperative suspects the method changes the mean weight, without specifying a direction; a two-tailed test ($H_1 : \mu \neq 38.0$) is therefore appropriate, since an increase or a decrease would both be of interest. Step 3 — Significance level: fix α in advance, conventionally $\alpha = 0.05$; it is the accepted probability of wrongly rejecting H_0. Step 4 — Test statistic: for a large sample, $Z = \dfrac{\bar{X} - \mu_0}{s/\sqrt{n}} \approx N(0,1)$ under H_0. Step 5 — Critical region: $|z| > 1.960$ at the 5% two-tailed level. Step 6 — Decision rule: reject H_0 if the computed z falls in the critical region, otherwise do not reject. Step 7 — Conclusion: interpret the decision in the context of the problem, in plain language. Mark scheme: hypotheses (1), justified two-tailed choice (2), role of α (1), statistic (1), critical region (1), decision rule (1), contextual conclusion requirement (1).

2. (6 marks) Compute:
$$z = \frac{39.1 - 38.0}{4.8/\sqrt{64}} = \frac{1.1}{0.6} \approx 1.83.$$
Since $|1.83| < 1.960$, we do not reject H_0. At the 5% level, the sample does not provide sufficient evidence that the new fermentation method changes the mean weight of dried beans per pod; the cooperative cannot yet claim any effect of the new method. Mark scheme: standard error 0.6 (1), $z = 1.83$ (2), comparison with 1.960 (1), decision (1), conclusion in context (1).

3. (4 marks) A Type II error here means concluding that the new method has no effect when in reality it does change the mean weight: the cooperative would miss a genuine improvement (or a genuine deterioration). Its probability β can be reduced by increasing the sample size (equivalently, reducing variability or accepting a larger α). Mark scheme: correct contextual meaning (2), valid method of reducing β (2).

11 SUMMARY SHEET



Mind map of the chapter Hypothesis Testing.

FICHE BILAN **L'essentiel à connaître pour l'examen**

Key definitions

- A **statistical hypothesis** is a claim about a population parameter (mean μ, proportion, variance).
- The **null hypothesis** H_0 is the statement of "no change / no difference"; it always contains an equality (e.g. $H_0 : \mu = \mu_0$).
- The **alternative hypothesis** H_1 is what we suspect instead: $\mu \neq \mu_0$ (two-tailed), $\mu > \mu_0$ or $\mu < \mu_0$ (one-tailed).
- The **significance level** α (often 5%, 1%, 10%) is the probability of rejecting H_0 when it is true.
- The **critical (rejection) region** is the set of values of the test statistic for which H_0 is rejected; its boundary values are the **critical values**.
- **Type I error**: reject H_0 when true; $P(\text{Type I}) = \alpha$. **Type II error**: accept H_0 when false; $P(\text{Type II}) = \beta$.

Formulas to know

- Test for a mean (σ known, or n large with $s \approx \sigma$):

$$z = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}}, \qquad \text{compare with } N(0,1).$$

- Difference of two means (large independent samples):

$$z = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\dfrac{\sigma_1^2}{n_1} + \dfrac{\sigma_2^2}{n_2}}} \quad \text{under } H_0 : \mu_1 = \mu_2.$$

- Critical values: two-tailed 5%: ±1.96; one-tailed 5%: 1.645; two-tailed 1%: ±2.576; one-tailed 1%: 2.326.
- Confidence interval link: $\bar{x} \pm z_{\alpha/2}\dfrac{\sigma}{\sqrt{n}}$; H_0 rejected iff μ_0 lies outside it.

The 5-step procedure

1. State H_0 and H_1 and the type of tail.
2. Choose the significance level α and find the critical value(s).
3. Compute the test statistic z from the sample.
4. Decide: reject H_0 if z falls in the critical region (or p-value $< \alpha$).
5. Conclude in the context of the problem, in plain language.

Common mistakes

- Writing H_1 with an equality sign, or using a two-tailed test when the question says "increase" or "decrease" (one-tailed).
- Forgetting to divide by $\sqrt{n}$: the standard error is $\sigma/\sqrt{n}$, not σ.
- Confusing α (Type I) with β (Type II); only α is fixed by the examiner.
- Saying "H_0 is proved" — we only fail to reject or reject; never prove.
- Wrong critical value: 1.96 is two-tailed 5%; for one-tailed 5% use 1.645.
- Not stating the conclusion in context ("the mean mass has changed" rather than just "reject H_0").

GCE tips

- Quote the formula, substitute values, then compute — method marks are given for correct substitution.
- Always draw a quick sketch of $N(0,1)$ with the critical region shaded.
- Keep 4 decimal places in z; compare carefully with the critical value.

- For "difference of means" questions, check samples are large ($n \geq 30$) so the normal approximation and $s \approx \sigma$ are valid.
- End every test with a sentence linking the decision to the real situation (e.g. a factory's bags of cement in Douala).

Signed • OnBuch+
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